

CHAPTER 1

INTRODUCTION

1.1 Background

With the rapid advancement of digital technology and artificial intelligence, traditional cultural services such as mehndi design selection are increasingly moving towards intelligent and automated solutions. Traditionally, customers choose mehndi designs by manually browsing through large collections of images or relying on the personal suggestions of artists, which can be time-consuming and often lacks personalization based on unique hand features such as palm size, finger length, hand shape, and spacing. During peak seasons like weddings and festivals, artists may find it difficult to provide individualized recommendations for every client, leading to inefficiencies and reduced service quality. Moreover, the mehndi industry still largely depends on informal communication methods and manual record-keeping for bookings, payments, and product sales, which can result in mismanagement, limited reach, and decreased customer satisfaction. The growing demand for digital convenience, personalized services, and efficient management systems highlights the need for technological integration in this domain. Recent developments in artificial intelligence, computer vision, and image processing have opened new possibilities for automating the analysis of hand images and generating suitable mehndi design recommendations. By leveraging these technologies, intelligent systems can identify geometric hand features, match them with predefined design attributes, and provide customized suggestions that enhance both aesthetic outcomes and user experience. The AI-Based Mehndi Design Recommendation System is therefore developed to modernize the traditional mehndi selection process by offering a smart, scalable, and user-friendly digital platform that benefits customers, supports artists in expanding their professional opportunities, and promotes innovation while preserving the cultural significance of mehndi art.

1.2 Motivation

The motivation behind developing the AI-Based Mehndi Design Recommendation System arises from the increasing need to modernize traditional artistic services through intelligent digital solutions. In contemporary society, customers expect personalized, fast, and convenient services that can be accessed through online platforms. However, the process of selecting suitable mehndi designs still largely depends on manual browsing and subjective

artist recommendations. This often results in time-consuming decision-making, lack of personalization, and difficulty in identifying designs that complement individual hand structures. As demand for customized and technology-driven services continues to grow, there is a clear need for an intelligent system that can automate and enhance the mehndi selection experience.

One of the major challenges faced by customers is the overwhelming number of mehndi design options available across various platforms. Users frequently spend significant time searching for designs that match their preferences, occasion, and hand features such as palm size, finger length, and hand shape. In many cases, customers may not have the expertise to determine which design style would best suit them, leading to dissatisfaction with the final outcome. This creates the necessity for a smart recommendation system that can analyze hand images and provide personalized design suggestions based on geometric and aesthetic compatibility.

Another important motivation for this project is the need to support mehndi artists in efficiently managing their services, bookings, and product sales. Many artists rely on informal communication channels or social media platforms, which may not provide structured management of customer interactions, design portfolios, or financial transactions. This can limit their professional growth and reduce operational efficiency. By integrating AI-driven recommendation capabilities with booking and management features, the system aims to create a unified digital environment that benefits both customers and artists.

Furthermore, the growing adoption of artificial intelligence, computer vision, and image processing technologies presents an opportunity to innovate within the cultural and creative industries. These technologies enable automated hand landmark detection, feature extraction, and intelligent design matching, making it possible to deliver accurate and personalized recommendations. The development of this system is therefore motivated by the need to improve user experience, enhance business scalability for artists, and introduce technological innovation while preserving the traditional significance and artistic value of mehndi design practices.

1.3 Scope & Objectives

The scope of this project focuses on developing an intelligent digital platform that automates the process of selecting suitable mehndi designs through artificial intelligence and image

processing techniques. The system is designed to support customers, mehndi artists, and administrators within a unified web-based environment that enhances service efficiency, personalization, and overall user experience. By integrating computer vision algorithms with traditional service management features such as booking and product sales, the platform aims to modernize the traditional mehndi industry while preserving its cultural significance. The system provides automated hand image analysis, personalized design recommendations, and structured management of artist services, thereby improving both customer satisfaction and professional productivity. The key objectives of the project are as follows:

1. To design and develop an AI-powered mehndi recommendation platform that enables users to upload plain hand images and receive personalized design suggestions within a single integrated system.
2. To implement computer vision and image processing techniques for automatic hand landmark detection and geometric feature extraction, enabling accurate matching between hand structure and suitable mehndi design styles.
3. To develop an intelligent recommendation mechanism that suggests the most appropriate mehndi designs based on features such as palm size, finger length, hand proportions, and design coverage preferences.

The system is intended to assist customers in selecting aesthetically suitable mehndi designs while helping artists efficiently manage bookings, showcase portfolios, and expand their professional reach. Although the platform does not replace the creativity and expertise of mehndi artists, it provides valuable technological support that enhances service delivery, promotes digital transformation in the cultural sector, and improves overall efficiency in the mehndi design selection process.

1.4 Problem Statement

The rapid advancement of digital technology and the growing demand for personalized services have significantly influenced various traditional industries, including the mehndi design sector. Customers today expect convenient, fast, and customized solutions when selecting mehndi designs for weddings, festivals, and special occasions. However, the process of choosing suitable designs still largely depends on manual browsing, artist suggestions, and subjective decision-making. While online platforms provide access to a wide range of design collections,

they often lack intelligent tools that can assist users in identifying designs that best match their hand structure and personal preferences.

One of the major challenges in the current mehndi selection process is the difficulty customers face in filtering through large volumes of design options. Users typically spend considerable time exploring different styles such as Arabic, bridal, or floral designs without clear guidance on which design would complement their hand shape, palm size, or finger proportions. This can lead to confusion, delayed decision-making, and dissatisfaction with the final outcome. The absence of automated recommendation systems that consider individual hand features highlights the need for intelligent technological solutions in this domain.

Another significant issue is the lack of structured digital management systems for mehndi artists and service providers. Many artists rely on informal communication methods such as phone calls or social media messages to handle bookings, manage portfolios, and promote their services. This can result in inefficiencies, scheduling conflicts, and limited professional visibility. Additionally, customers may face difficulties in identifying reliable artists, comparing services, or ensuring transparency in pricing and quality.

Furthermore, existing digital platforms in the mehndi industry rarely incorporate advanced technologies such as artificial intelligence or computer vision to enhance user experience. The absence of automated hand image analysis, personalized design matching, and integrated booking management systems reduces operational efficiency and restricts business scalability. Therefore, there is a clear need for an intelligent, integrated platform that can automate the mehndi design recommendation process, improve service management, and provide a more efficient and user-friendly experience for both customers and artists.

CHAPTER 2

REQUIREMENTS AND ANALYSIS

2.1 Existing System & Problem Statement

With the growing influence of digital technology and online service platforms, traditional creative industries such as the mehndi design sector are gradually transitioning towards digital environments. Customers increasingly rely on online sources such as social media, design galleries, and artist portfolios to select suitable mehndi designs for various occasions. While this digital shift has improved access to a wide range of design options, it has also introduced challenges related to personalization, service management, and decision-making efficiency.

Existing mehndi selection systems mainly focus on displaying design collections through websites or social media platforms. These platforms allow customers to view different styles such as Arabic, bridal, or floral designs, and contact artists for booking services. However, such systems lack intelligent mechanisms that can assist users in identifying designs that best suit their individual hand features, preferences, or event requirements. As a result, customers often spend considerable time browsing numerous designs without clear guidance, leading to confusion and reduced satisfaction with the final outcome.

One of the major limitations of the current system is the absence of automated recommendation tools based on hand structure analysis. Traditional methods depend on subjective suggestions from artists, which may vary depending on experience and workload. Without technological support, it becomes difficult to ensure accurate personalization for each customer. Additionally, the manual process of selecting designs can be inefficient, especially during peak seasons such as weddings and festivals when artists manage multiple clients simultaneously.

Another limitation is related to the lack of structured digital management systems for artist services, bookings, and product sales. Many artists rely on informal communication methods such as phone calls or messaging applications to handle client interactions, maintain design portfolios, and manage payments. This unorganized approach can lead to scheduling conflicts, limited professional visibility, and inefficient business operations. Customers may also face challenges in identifying trustworthy artists, comparing services, or ensuring transparency in pricing and quality.

Furthermore, most existing platforms in the mehndi domain do not utilize advanced technologies such as artificial intelligence, computer vision, or image processing to enhance

user experience. The absence of automated hand image analysis and intelligent design matching reduces operational efficiency and restricts the scalability of digital mehndi services. Therefore, the problem lies in the lack of an integrated, intelligent platform that can automate the mehndi design recommendation process while supporting efficient service management. A more advanced solution is required that incorporates artificial intelligence techniques for hand feature analysis, personalized design suggestions, and streamlined booking management to improve overall efficiency and user satisfaction in the mehndi design industry.

2.2 Solution Overview

To overcome the limitations of traditional mehndi design selection methods, the proposed **AI-Based Mehndi Design Recommendation System** introduces an integrated digital solution that combines intelligent image analysis with service management functionalities. The system is designed to support customers, mehndi artists, and administrators through a unified web-based platform that enhances personalization, operational efficiency, and user experience. By incorporating artificial intelligence and computer vision techniques, the platform modernizes the traditional mehndi selection process and provides structured digital support for creative services.

The proposed system enables customers to upload plain hand images and receive personalized mehndi design recommendations through a centralized interface. Mehndi artists can showcase their design portfolios, manage service bookings, and sell henna-related products within the same platform. Administrators oversee the system by managing user registrations, approving designs, monitoring transactions, and ensuring smooth platform operation. This integrated approach ensures effective coordination among all stakeholders while promoting digital transformation in the mehndi industry.

The first stage of the system focuses on automated hand image processing and feature extraction. When a customer uploads a hand image, the system preprocesses the image to remove noise, segment the hand region, and detect key landmarks such as fingertips, palm boundaries, and wrist points. Using computer vision algorithms, geometric features including palm width, finger length, and hand proportions are extracted. These features are then analyzed to determine the most suitable mehndi design styles for the user.

Following feature extraction, the system performs intelligent design matching using predefined design attributes stored in the database. Based on the similarity between hand features and design characteristics, the platform recommends the top matching mehndi styles such as Arabic, bridal, or floral designs. Customers can review the suggested designs, compare artist portfolios, and proceed with booking services directly through the platform. This automated recommendation mechanism significantly reduces the time required for design selection and improves personalization accuracy.

In addition to recommendation capabilities, the system provides structured management tools for artist services, bookings, and product sales. Artists can upload new designs, update pricing, manage customer requests, and track their earnings through an organized dashboard. The platform also supports secure online payment processing and feedback collection, enabling continuous service improvement. By integrating intelligent recommendation, service management, and digital transaction features within a single system, the proposed solution enhances convenience, transparency, and scalability in the modern mehndi design ecosystem.

2.3 Feasibility Study

A feasibility study evaluates whether the proposed system can be successfully developed and implemented. It examines factors such as technical, economic, and operational feasibility to ensure the practicality of the system.

2.3.1 Technical Feasibility

Technical feasibility focuses on determining whether the required technologies, tools, and resources are available to develop the proposed system. The AI-Based Mehndi Design Recommendation System can be implemented using widely available web technologies and artificial intelligence frameworks that support image processing and recommendation functionalities.

The backend of the system can be developed using the Python programming language and the Django web framework, which provides a reliable environment for building secure and scalable web applications. Image processing and hand landmark detection can be implemented using computer vision libraries such as OpenCV and MediaPipe, which enable the system to detect

hand features and extract important measurements such as palm width, finger length, and hand proportions.

For the frontend interface, technologies such as HTML, CSS, JavaScript, and Bootstrap can be used to design a responsive and interactive user interface that allows customers to upload hand images, view recommended designs, and book artists. The system database can be managed using MySQL or SQLite to store user information, mehndi designs, booking details, and product records.

Since these technologies are widely used, well-documented, and supported by active developer communities, the technical resources required for implementing the system are readily available. Therefore, the proposed system is technically feasible.

2.3.2 Economic Feasibility

Economic feasibility evaluates the financial cost involved in developing and maintaining the proposed system. The AI-Based Mehndi Design Recommendation System can be developed using open-source technologies and freely available development tools, which significantly reduces the overall cost of implementation.

Programming languages such as Python, frameworks like Django, and libraries including OpenCV and MediaPipe are open-source and do not require any licensing fees. Similarly, database systems such as MySQL or SQLite are freely available for development purposes. Development environments like PyCharm Community Edition or Visual Studio Code can also be used without additional cost.

The system does not require expensive hardware infrastructure for implementation. A standard computer with moderate processing capability, basic storage capacity, and internet connectivity is sufficient to develop and run the application. Because the project relies on open-source software and standard hardware resources, the overall development and maintenance costs remain low. This makes the system economically feasible for educational projects and small-scale commercial deployment.

2.3.3 Operational Feasibility

Operational feasibility evaluates how effectively the proposed system will function in real-world conditions and how easily users can interact with it. The AI-Based Mehndi Design Recommendation System is designed to be user-friendly and accessible for different types of users including customers, mehndi artists, and administrators.

Customers can easily upload a plain hand image through the web interface and receive recommended mehndi designs based on their hand features. They can browse different design styles, view artist portfolios, book mehndi services, and purchase henna products through the platform. Mehndi artists can upload their design collections, manage bookings, track product orders, and monitor their earnings through a dedicated dashboard.

Administrators manage the overall system operations by approving designs, monitoring transactions, managing users, and analyzing system activities. Most system functions such as image processing, design recommendation, and booking management are automated, which reduces manual effort and improves efficiency.

Since the platform provides simple navigation, clear instructions, and an intuitive user interface, it can be easily adopted by users with basic technical knowledge. Therefore, the proposed system is operationally feasible and suitable for real-world implementation in the mehndi service industry.

CHAPTER 3

LITERATURE REVIEW

3.1 Computer Vision Techniques for Hand Landmark Detection

[Zhang, Cao, IEEE Conference on Computer Vision, 2020]

Computer vision technologies play a crucial role in modern image analysis applications, particularly in detecting and interpreting human hand structures. Zhang and Cao conducted extensive research on automated hand landmark detection using deep learning-based approaches. Their study focused on identifying important hand points such as fingertips, joints, palm centers, and wrist boundaries from digital images captured under varying environmental conditions.

The research demonstrated that convolutional neural networks (CNNs) and deep learning frameworks significantly improve the accuracy of hand feature detection compared to traditional image processing techniques. Accurate hand landmark detection enables further geometric analysis, including palm width measurement, finger length estimation, and hand shape classification. These features are essential for applications requiring personalization, such as biometric authentication, gesture recognition, and aesthetic design recommendation systems.

In addition, the study highlighted that integrating computer vision models into real-time applications enhances user interaction and automation. When applied to creative industries like mehndi design, automated hand landmark detection allows systems to analyze hand geometry and recommend appropriate design styles. This reduces manual effort for artists and improves customer satisfaction by providing personalized suggestions.

Furthermore, digital platforms that integrate image analysis with service booking features can streamline workflow for both customers and artists. Once hand features are extracted, users can be guided toward suitable design options, artist portfolios, and product offerings. This integration of visual intelligence with service management supports efficient design selection and improves digital transformation in traditional art services.

3.2 Artificial Intelligence in Recommendation Systems**[Adomavicius, Tuzhilin, IEEE Transactions on Knowledge and Data Engineering, 2005]**

Artificial intelligence-based recommendation systems have become an integral part of modern digital platforms that provide personalized user experiences. Adomavicius and Tuzhilin explored various recommendation methodologies, including collaborative filtering, content-based filtering, and hybrid recommendation models. These systems analyze user preferences, behavioral data, and contextual information to generate relevant suggestions.

The research highlighted that intelligent recommendation systems improve decision-making processes by reducing the time users spend searching for suitable options. In design-oriented

applications, AI-based recommendation engines can analyze visual features, user preferences, and historical interaction data to provide customized suggestions. This capability is particularly valuable in artistic domains where personalization plays a significant role in customer satisfaction.

Additionally, recommendation systems are widely used in e-commerce platforms to suggest products based on customer behavior. Integrating similar techniques in mehndi service platforms enables users to receive personalized design recommendations along with related product suggestions such as henna cones, aftercare products, or decorative accessories. This enhances the overall user experience and promotes digital commerce in creative industries.

The study also emphasized the importance of integrating recommendation systems with service management modules such as booking, payment processing, and feedback collection. By combining intelligent design matching with online transaction systems, digital platforms can provide seamless end-to-end user experiences, improving both service efficiency and business scalability.

3.3 Image Processing Applications in Cultural Art Digitization

[Smith, Cultural Heritage Computing Journal, 2019]

The digitization of traditional art forms has gained significant attention with the advancement of image processing and artificial intelligence technologies. Smith investigated how digital transformation helps preserve cultural heritage while enabling innovative applications in artistic domains. The study focused on converting traditional artistic patterns into digital formats for automated analysis, classification, and recommendation.

Digital image processing techniques enable accurate extraction of visual features such as pattern density, motif distribution, and structural complexity in artistic designs. In the context of mehndi art, these techniques can be used to categorize designs into different styles such as Arabic, bridal, floral, or minimalist patterns. Automated classification of designs simplifies the process of organizing digital design libraries and improves accessibility for users.

Furthermore, integrating cultural art digitization with online platforms allows artists to showcase their work to a broader audience. Digital portfolios combined with intelligent search and recommendation tools enhance visibility and professional opportunities for artists. The study also emphasized the role of digital platforms in facilitating online booking and secure payment systems, enabling artists to monetize their skills effectively.

By combining image processing technologies with digital commerce features, cultural art industries can achieve sustainable growth while preserving traditional artistic values. Such

integration supports innovation, enhances customer engagement, and promotes efficient management of artistic services and product sales.

3.4 Hand Geometry Analysis for Personalized Design Systems

[Kumar, Jain, International Journal of Pattern Recognition, 2018]

Hand geometry analysis is widely applied in biometric systems, ergonomic product design, and personalized aesthetic applications. Kumar and Jain proposed methods for extracting geometric measurements such as palm width, finger length ratios, hand area, and spatial relationships between fingers. These measurements help categorize individuals into structural groups for personalization purposes.

The study demonstrated that accurate hand geometry analysis improves the effectiveness of recommendation systems in applications requiring customization. For instance, in jewelry fitting or glove manufacturing, personalized measurements ensure comfort and visual compatibility. Similarly, in artistic applications such as mehndi design, hand feature analysis can guide the selection of suitable patterns that complement individual hand structures.

The research also highlighted the importance of integrating geometric analysis with digital service platforms. Once user-specific measurements are obtained, the system can automatically suggest appropriate design styles and enable users to book professional services. Additionally, integrating product sales modules allows customers to purchase related items such as henna products, decorative accessories, or skincare solutions.

By combining biometric analysis with digital service management, platforms can provide comprehensive solutions that enhance personalization, streamline service delivery, and support business scalability for artists and service providers.

3.5 Machine Learning Techniques in Image-Based Classification

[Goodfellow, Bengio, Courville, Deep Learning Book, MIT Press]

Machine learning techniques have revolutionized image classification and pattern recognition tasks across multiple domains. Goodfellow et al. explained how deep learning architectures such as convolutional neural networks (CNNs) learn hierarchical visual features from large datasets. These models can identify complex patterns, classify images, and predict outcomes with high accuracy.

The research highlighted that machine learning-based image classification systems are particularly effective in applications involving artistic design recognition. In the context of mehndi design recommendation, deep learning models can analyze visual characteristics such

as pattern complexity, coverage area, and motif structure to categorize designs into different style groups.

Furthermore, integrating machine learning models with digital platforms enables automation in service recommendation and product suggestion processes. Users can receive personalized design recommendations based on both hand features and design characteristics. The platform can also suggest related products and facilitate online booking and payment processes, creating a complete digital ecosystem.

The study concluded that combining machine learning techniques with user-centric digital platforms enhances automation, improves personalization accuracy, and supports efficient management of creative services and e-commerce activities.

3.6 Digital Service Booking Platforms in Creative Industries

[Lee, Service Computing Journal, 2017]

Digital service booking platforms have significantly transformed the operational models of creative and service-based industries. Lee analyzed the impact of online booking systems on service efficiency, customer satisfaction, and business growth. The study highlighted that traditional service industries often rely on manual appointment scheduling and informal communication, which can lead to inefficiencies, scheduling conflicts, and reduced customer engagement.

The research emphasized that online booking platforms provide structured mechanisms for managing service requests, appointment scheduling, and customer communication. These systems allow customers to view available services, compare options, and book appointments conveniently through digital interfaces. In creative domains such as mehndi artistry, digital booking platforms enable artists to manage multiple clients effectively while reducing administrative workload.

Additionally, the integration of real-time notifications and automated reminders enhances transparency and improves customer trust. The study also explored the importance of integrating online payment gateways into booking systems, enabling secure financial transactions and reducing reliance on manual payment methods. By combining service booking, payment processing, and feedback management features, digital platforms can create comprehensive service ecosystems that improve operational efficiency and business scalability.

3.7 E-commerce Integration in Artistic Service Platforms

[Porter, Harvard Business Review, 2018]

The integration of e-commerce technologies into artistic service platforms has created new opportunities for artists and designers to expand their professional reach. Porter discussed how digital marketplaces enable creative professionals to sell products, manage inventory, and interact with customers through centralized platforms. The research highlighted that combining artistic services with product sales enhances revenue generation and supports sustainable business models.

In the context of mehndi artistry, artists can sell related products such as henna cones, aftercare kits, decorative accessories, and design templates through digital platforms. E-commerce features such as product categorization, pricing management, and order tracking improve operational efficiency and provide customers with convenient purchasing options.

The study also emphasized the importance of integrating secure online payment systems to facilitate seamless financial transactions. Digital payment gateways support multiple payment methods, including credit cards, digital wallets, and online banking, ensuring flexibility and security for users. By combining service booking, product sales, and payment processing within a single platform, artistic service providers can create comprehensive digital ecosystems that enhance customer engagement and business growth.

3.8 Human-Computer Interaction in Visual Recommendation Systems

[Nielsen, Usability Engineering Journal, 1994]

Human-computer interaction (HCI) plays a crucial role in the design and adoption of digital recommendation systems. Nielsen explored usability principles that improve user experience in software applications, particularly those involving visual content and interactive features. The study emphasized that intuitive interface design, clear navigation structures, and responsive layouts are essential for ensuring user satisfaction and system effectiveness.

In visual recommendation platforms such as mehndi design systems, effective HCI design enables users to easily upload images, browse recommended designs, and complete service bookings. Interactive visualization tools allow users to preview designs and compare different styles before making decisions. This enhances engagement and reduces the cognitive effort required for decision-making.

The research also highlighted that integrating feedback mechanisms and user support features improves system reliability and continuous improvement. By combining usability principles with intelligent recommendation technologies, digital platforms can provide seamless user experiences that encourage adoption and long-term engagement.

3.9 Artificial Intelligence-Based Pattern Recognition in Artistic Designs

[Chen, Pattern Recognition Letters, 2021]

Artificial intelligence-based pattern recognition techniques have been widely applied in analyzing artistic designs and decorative motifs. Chen proposed machine learning approaches for identifying structural and stylistic characteristics in visual patterns. The research demonstrated that AI models can classify artistic designs based on visual similarity, pattern density, and structural complexity.

In mehndi design applications, pattern recognition algorithms can categorize designs into different style groups such as Arabic, bridal, contemporary, or minimalist patterns. Automated classification improves the organization of digital design libraries and enhances the accuracy of recommendation systems.

The study also emphasized the role of AI in supporting digital marketing and product recommendation strategies. By analyzing user preferences and interaction patterns, AI systems can suggest related products and services, improving cross-selling opportunities. Integrating pattern recognition technologies with online booking and payment systems enables the creation of intelligent platforms that automate multiple aspects of artistic service delivery.

3.10 Digital Transformation in Traditional Art and Cultural Industries**[UNESCO Cultural Innovation Report, 2022]**

Digital transformation has become a critical factor in preserving and promoting traditional art forms in modern society. The UNESCO Cultural Innovation Report examined how digital technologies such as artificial intelligence, online platforms, and digital commerce tools contribute to the sustainability of cultural industries. The study highlighted that digital platforms enable artists to reach global audiences and adapt to changing consumer preferences. The research emphasized that integrating AI-driven recommendation systems with online service management features enhances the accessibility and efficiency of traditional artistic services. Digital platforms allow customers to discover suitable designs, book professional services, and purchase related products through unified interfaces.

Furthermore, the adoption of secure online payment systems supports financial transparency and reduces transaction-related challenges. Digital transformation also facilitates data-driven decision-making, enabling artists and administrators to analyze customer behavior, service demand, and product performance. The study concluded that technology-driven innovation is essential for ensuring the long-term sustainability and growth of traditional art industries in the digital era.

CHAPTER 4

DIAGRAMS

System diagrams play a vital role in software design as they provide a clear and structured representation of the architecture, workflow, and interaction between different components of a system. These diagrams help in visualizing how the AI-Based Mehndi Design Recommendation System operates internally and how various users interact with the platform. By using diagrams such as Data Flow Diagrams (DFD), Use Case Diagrams, Class Diagrams, Activity Diagrams, Sequence Diagrams, and System Architecture Diagrams, both developers and users can easily understand the flow of information and the responsibilities of each module within the system.

The diagrams included in this chapter illustrate the overall structure and working of the proposed mehndi recommendation system. They demonstrate how different users such as customers, mehndi artists, and administrators interact with the platform. The diagrams also explain how customers upload hand images, how the system processes these images using artificial intelligence techniques, and how personalized mehndi design recommendations are

generated. In addition, they show how users can browse designs, book mehndi artists, and purchase henna products through the platform.

Furthermore, these diagrams highlight the internal modules of the system, including image preprocessing, hand landmark detection, feature extraction, and the recommendation engine. They also represent how data flows between different components such as the user interface, processing modules, and the database where information like user details, designs, bookings, and product data are stored.

Each diagram presents a different perspective of the system, ranging from high-level user interactions to detailed internal processes. Together, these visual representations simplify the understanding of system functionality, improve communication between developers and stakeholders, and provide a strong foundation for the design and implementation of the AI-Based Mehndi Design Recommendation System.

4.1 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation used to illustrate how data moves through a system. It shows how information enters the system, how it is processed by different modules, and how the processed data is stored or delivered as output. DFDs help developers and system designers clearly understand the logical flow of information and the interactions between different components of the system.

In the AI-Based Mehndi Design Recommendation System, the DFD explains how data flows between users and system modules. It demonstrates how customers, mehndi artists, and administrators interact with the platform. For example, customers can upload plain hand images, view AI-recommended mehndi designs, book artists, and purchase henna products. Mehndi artists can upload their design portfolios, manage bookings, and handle product sales, while administrators manage users, approve designs, and monitor overall system activities.

The DFD also highlights the major components of the system, including:

- System processes (image processing, feature extraction, recommendation engine)
- Data stores (user data, mehndi designs, bookings, product details)

- Data flow between components (input images, processed data, recommendations, transactions)

The purpose of using a DFD is to simplify the understanding of system operations and clearly represent how data is processed and transferred within the platform. It helps in identifying how different modules interact, ensures efficient data management, and provides a structured view of the overall workflow of the AI-Based Mehndi Design Recommendation System.

4.1.1 DFD Level 0

DFD Level 0, also known as the context diagram, represents the entire system as a single process and shows how the system interacts with external entities. It provides a high-level overview of the system without going into detailed internal processes. The main purpose of this diagram is to show the interaction between users and the system. In the AI-Based Mehndi Design Recommendation System, the primary external entities include Customers (Users), Mehndi Artists, and Administrators who interact with the system to perform various activities related to design selection, booking, and management. Customers use the platform to register, log in, upload plain hand images, view AI-recommended mehndi designs, book mehndi artists, and purchase henna products. Mehndi artists interact with the system to upload mehndi design portfolios, manage booking requests, update product details, and track their earnings. Administrators manage users, approve designs and products, monitor system activities, and handle feedback or complaints. The users provide input data to the system such as registration details, hand images, design uploads, booking requests, and product information. The system processes this data using artificial intelligence and image processing techniques and stores it in the database. After processing, the system generates outputs such as recommended mehndi designs, booking confirmations, product order details, payment transactions, feedback responses, and system notifications. Thus, the Level 0 Data Flow Diagram presents the AI-Based Mehndi Design Recommendation System as a single process that connects external users with the internal system functions and database.

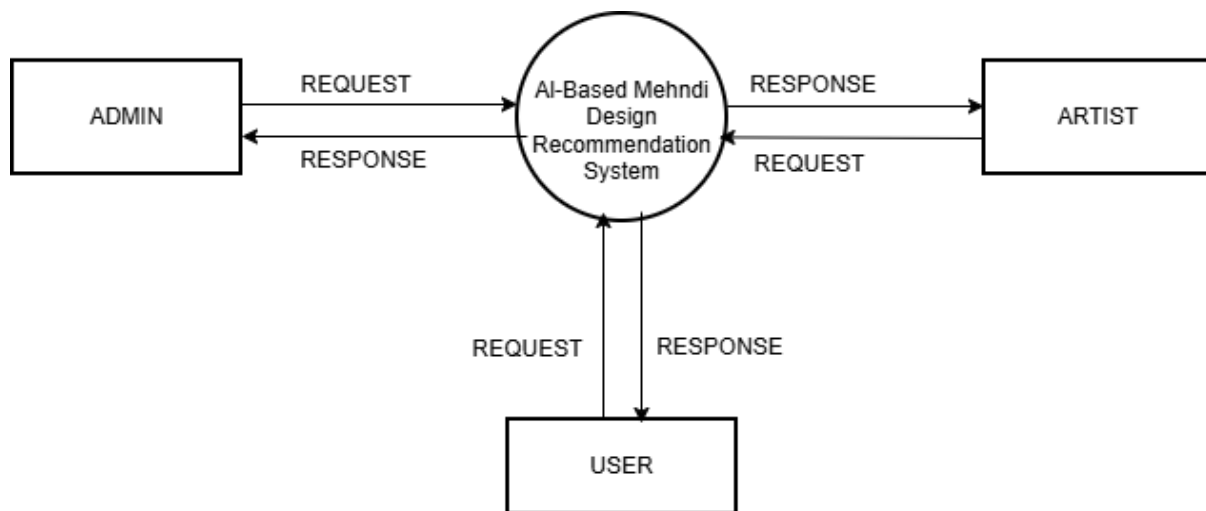


Figure 4.1.1 DFD Level 0

Components of the Level 0 DFD

External Entities

Admin: The administrator is responsible for managing the overall functioning of the system. The admin can manage customers and mehndi artists, approve uploaded designs and products, monitor bookings and transactions, and handle complaints and feedback. The administrator ensures that the system operates smoothly and efficiently.

Mehndi Artist: Mehndi artists interact with the system by uploading their design collections, tagging design types, managing booking requests, and selling henna products. They can also track their earnings from both services and product sales, making the platform useful for business growth.

User(Customer): Customers use the system to access personalized mehndi services. They can register and log in to the platform, upload hand images for AI-based recommendations, browse designs, book artists, purchase products, and provide feedback. The system helps users select suitable designs easily and efficiently.

4.1.2 DFD Level 1

DFD Level 1 provides a more detailed view of the system by expanding the Level 0 diagram into several smaller processes. It explains how the main EduTrack system is divided into different functional modules and how data flows between these modules and the users. This level shows how requests from users are processed and how the system generates appropriate responses.

DFD Level 1 – Admin

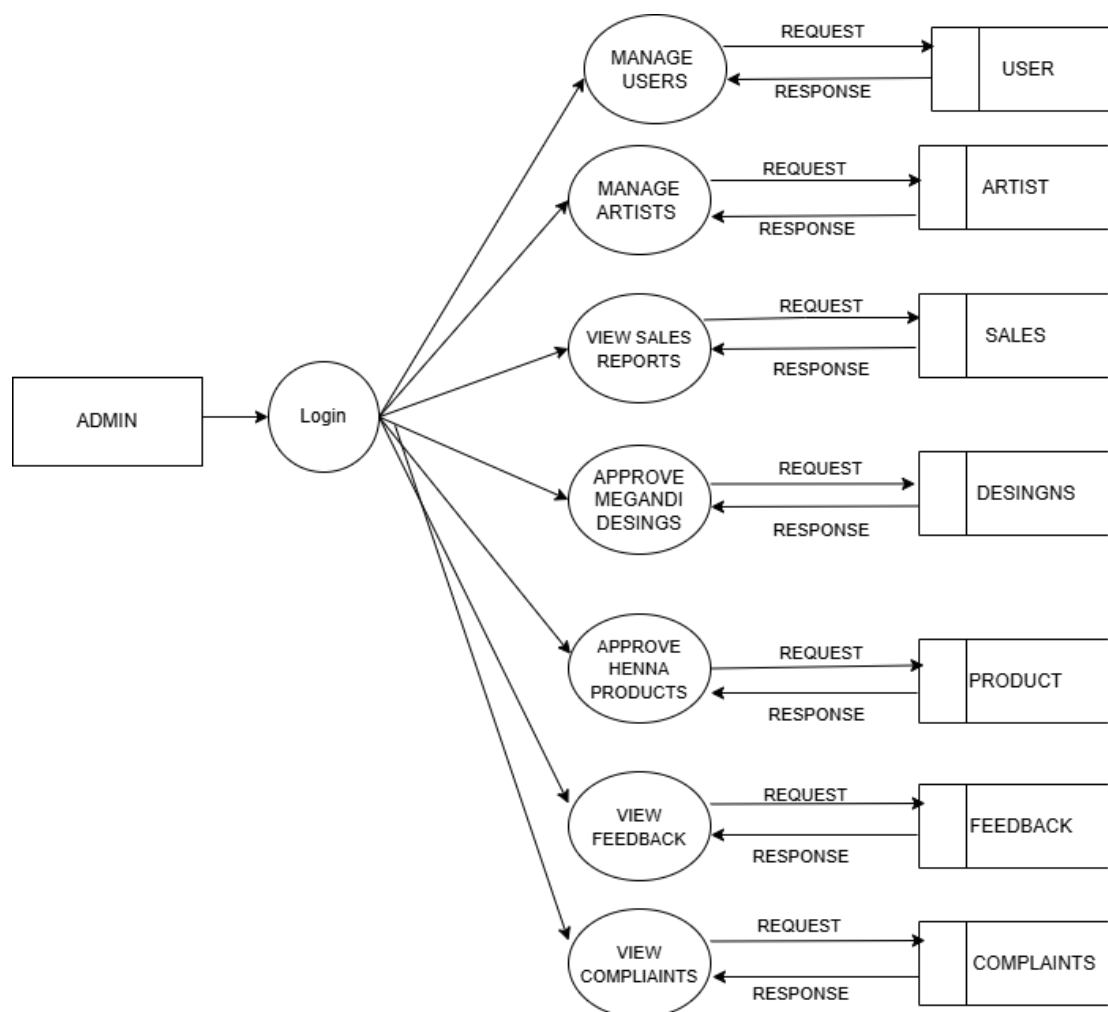


Figure 4.1.2 Admin DFD Level 1

DFD Level provides a detailed representation of the system by breaking down the main process shown in Level 0 into multiple sub-processes. It illustrates how data flows between internal modules such as user management, image processing, design recommendation, booking, product management, and payment processing. In the AI-Based Mehndi Design Recommendation System, the process begins with user authentication where customers, artists,

and admins log in to the system. Customers can upload hand images, which are processed through image preprocessing, hand landmark detection, and feature extraction modules. The extracted features are then sent to the recommendation engine, which generates suitable mehndi design suggestions based on stored design data. After receiving recommendations, customers can select designs, book mehndi artists, and purchase henna products through the platform. The booking module manages appointment scheduling and stores booking details, while the product module handles product listings and orders. The payment module ensures secure online transactions for both services and product purchases. Mehndi artists manage design uploads, bookings, and product sales, while administrators monitor system activities, approve content, and manage users. All data is stored and retrieved from databases such as user data, design records, bookings, products, and payments, ensuring smooth system operation and efficient data flow between all components.

DFD Level 1 – Artist

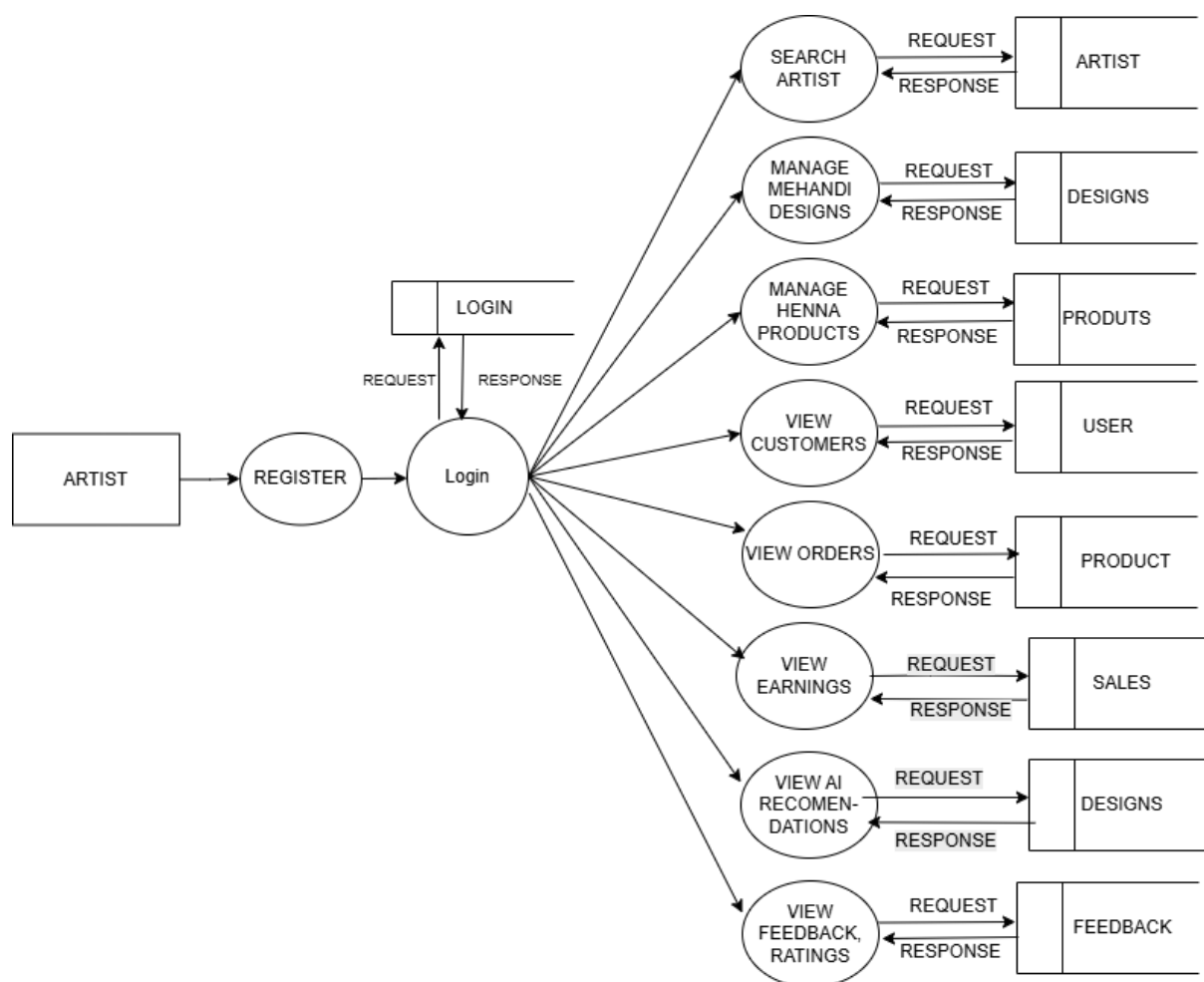


Figure 4.1.3 Artist DFD Level 1

The Artist module handles the design and service management within the AI-Based Mehndi Design Recommendation System. Mehndi artists register in the system and log in after receiving approval from the administrator. Once logged in, artists can upload mehndi designs, categorize them based on style (such as Arabic, bridal, or floral), and manage their design portfolio for customers to view.

Artists can also manage booking requests from customers, accept or reject appointments, and schedule their services accordingly. In addition, they can add henna products for sale, update pricing, and manage product availability. The system stores all design details, booking information, and product data in the database and provides access to customers. Artists can also view customer feedback, track their earnings from both bookings and product sales, and respond to customer queries through the system.

DFD Level 1 – User (Customer)

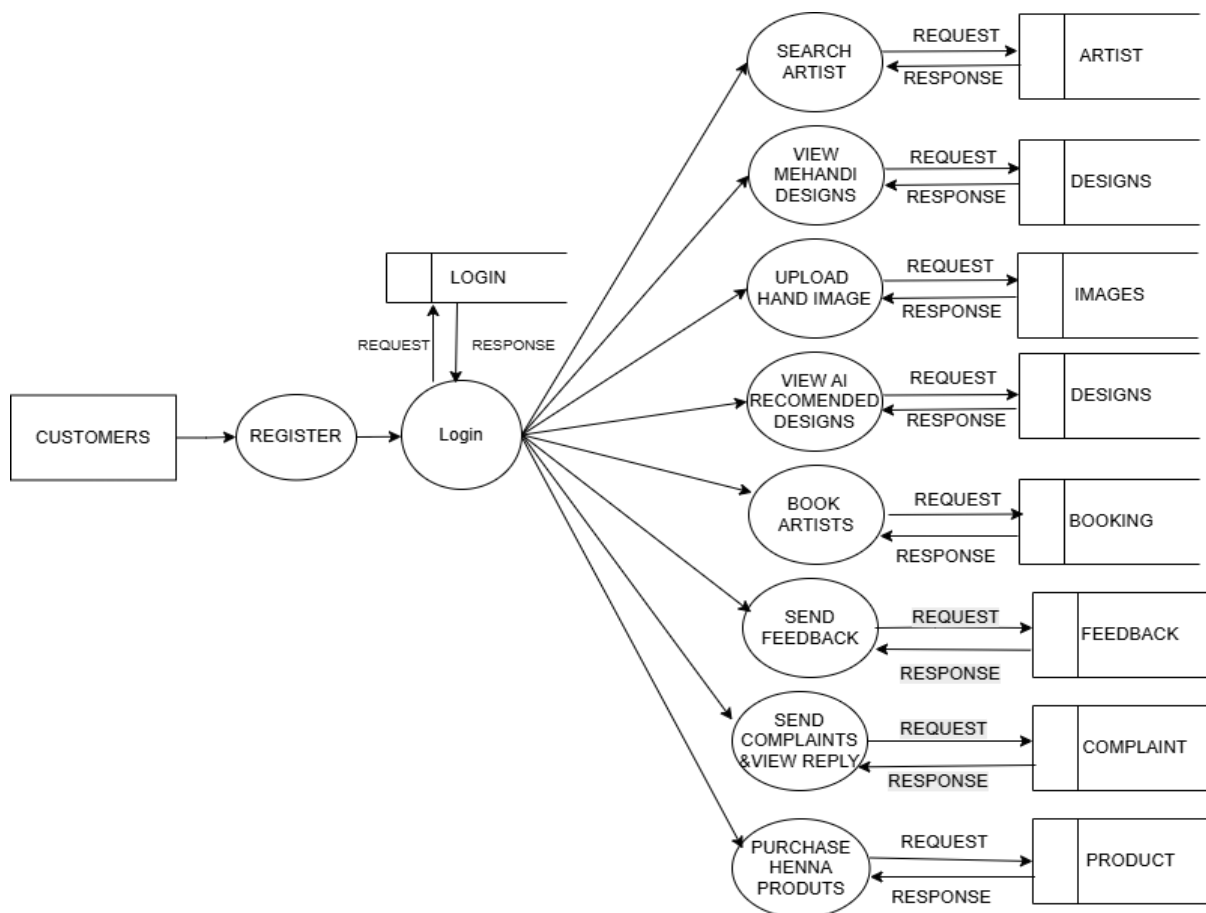


Figure 4.1.4 User(customer) DFD Level 1

The Customer module handles the user interaction and service utilization within the AI-Based Mehndi Design Recommendation System. Customers first register in the system and log in using their credentials to access various features. After successful authentication, users can navigate through the platform easily using a user-friendly interface designed for smooth interaction.

Once logged in, customers can upload a plain hand image to the system. The uploaded image is processed through multiple stages such as image preprocessing, hand landmark detection, and feature extraction. Based on the extracted hand features like palm size and finger length, the system generates personalized mehndi design recommendations that best suit the user's hand structure and preferences.

After receiving the recommendations, customers can browse different mehndi designs, compare styles, and select their preferred design. They can also view available mehndi artists, check their design portfolios, and send booking requests for services. The system allows users to schedule appointments based on availability, making the booking process simple and efficient.

In addition to service booking, customers can purchase henna products listed by artists and complete secure online payments through the platform. All activities such as image uploads, bookings, product purchases, and feedback are stored in the database. Customers can also track their booking status, view order history, and provide ratings or feedback, ensuring a complete and user-friendly experience.

4.2 Use Case Diagram

A Use Case Diagram illustrates the interaction between users and the system. It shows the different actions that users can perform and how the system responds to those actions. This diagram helps identify the main functionalities of the system and the roles of different actors involved. In the AI-Based Mehndi Design Recommendation System, the main actors are Admin, Mehndi Artist, and Customer. Each actor interacts with the system through a web interface to perform specific tasks related to mehndi design recommendation, booking services, product purchase, and system management.

Customers use the system to register, log in, upload plain hand images, receive AI-based mehndi design recommendations, browse designs, book mehndi artists, purchase henna products, and provide feedback. Mehndi artists interact with the system to upload designs, manage portfolios, accept or reject booking requests, add products for sale, and track earnings. Administrators control the overall system by managing users, approving artist registrations, approving designs and products, handling feedback and complaints, and monitoring system activities. The Use Case Diagram helps to clearly represent all the functionalities available within the AI-Based Mehndi Design Recommendation System and shows how each actor interacts with the platform. It provides a high-level overview of system behavior and helps developers understand the relationship between different system operations and user roles.

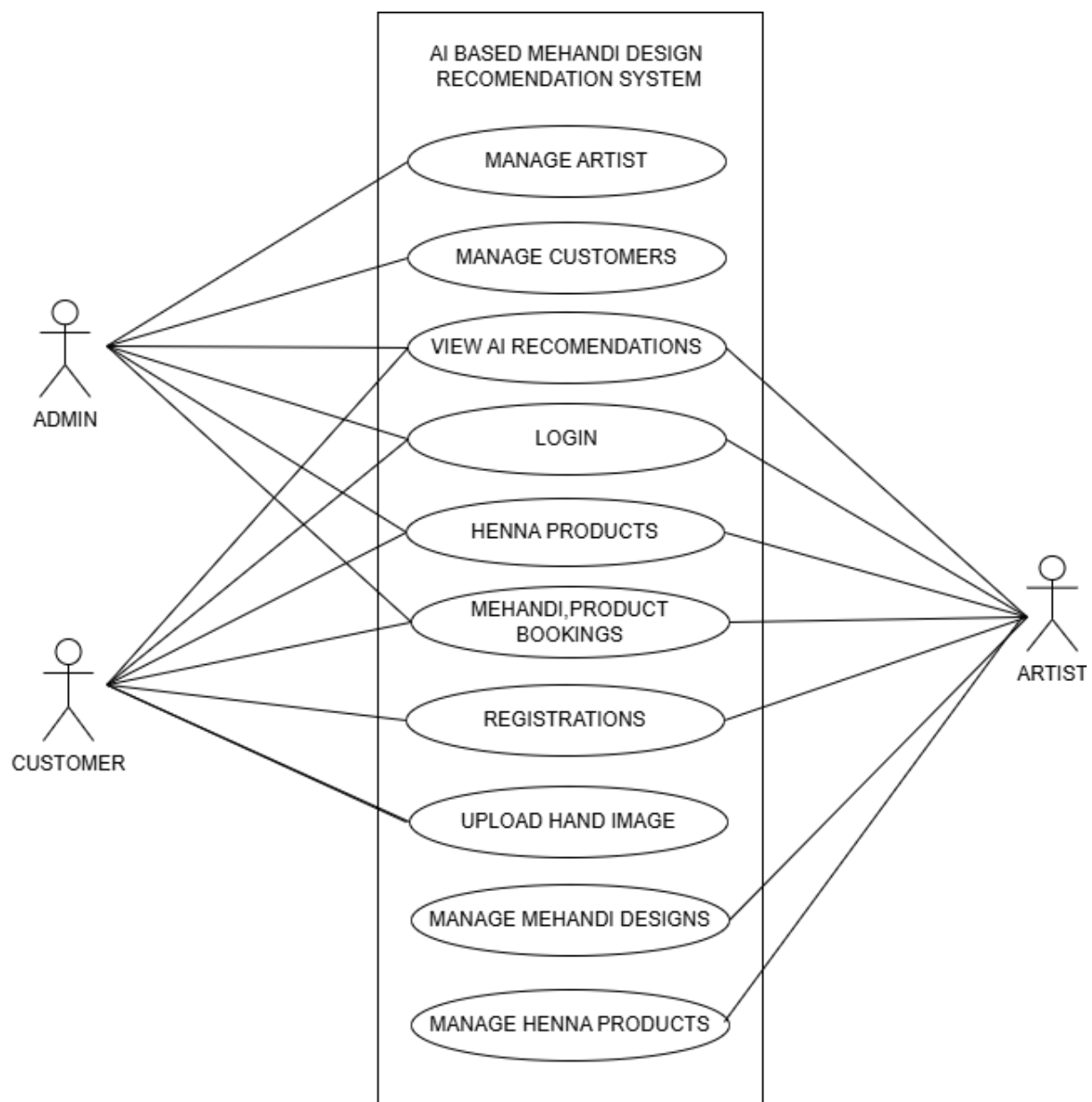


Figure 4.2.1 Use case diagram

Actors:

Admin: The administrator manages the overall functioning of the system. The admin controls customer and artist accounts, approves artist registrations, verifies uploaded designs and products, manages complaints and feedback, and monitors system activities.

Mehndi Artist: Mehndi artists use the system to manage their design portfolio and services. They can register, log in, upload mehndi designs, manage bookings, add henna products, update prices, and view earnings from both bookings and product sales.

Customer (User): Customers interact with the system to access mehndi-related services. They can register, log in, upload hand images, view AI-recommended mehndi designs, browse artist profiles, book services, purchase products, and give ratings or feedback.

Use Cases:

Register: Users can create an account in the system by providing personal details. The system stores this information in the database and allows users to access the platform after registration.

Login: Users log in to the system using their credentials. After successful authentication, the user can access system features based on their role.

Upload Hand Image: Customers can upload a plain hand image to the system. The image is processed by the AI module to analyze hand features and generate suitable mehndi design recommendations.

View Recommended Designs: Customers can view mehndi designs recommended by the system based on the uploaded hand image and extracted hand features.

Upload Designs: Mehndi artists can upload new mehndi designs to the platform. These designs are stored in the database and made available for customer browsing after admin approval.

Book Artist: Customers can select a mehndi artist and send a booking request for service. The system stores booking details and notifies the artist.

Manage Bookings: Artists can accept, reject, or update booking requests received from customers. Booking details are maintained within the system.

Add Products: Artists can add henna products such as cones or related items for sale. Product details, pricing, and availability are stored in the database.

Purchase Products: Customers can browse and purchase henna products available on the platform. The system processes the order and stores transaction details.

Online Payment: Customers can make secure online payments for both service bookings and product purchases. The payment details are recorded in the system.

Manage Users: The administrator can manage artist and customer accounts, including approvals, monitoring activities, and maintaining system security.

Provide Feedback / Complaint: Customers and artists can send feedback or complaints to the administrator, and the admin can review and respond to them through the system.

4.3 Class Diagram

The Class Diagram represents the structural design of the AI-Based Mehndi Design Recommendation System. It shows the different classes in the system, their attributes, methods, and the relationships between them. This diagram helps developers understand how the system components are organized and how they interact with each other. In the mehndi recommendation system, several classes represent the main entities involved in managing user activities, artist services, mehndi designs, bookings, products, administration, and the AI recommendation process.

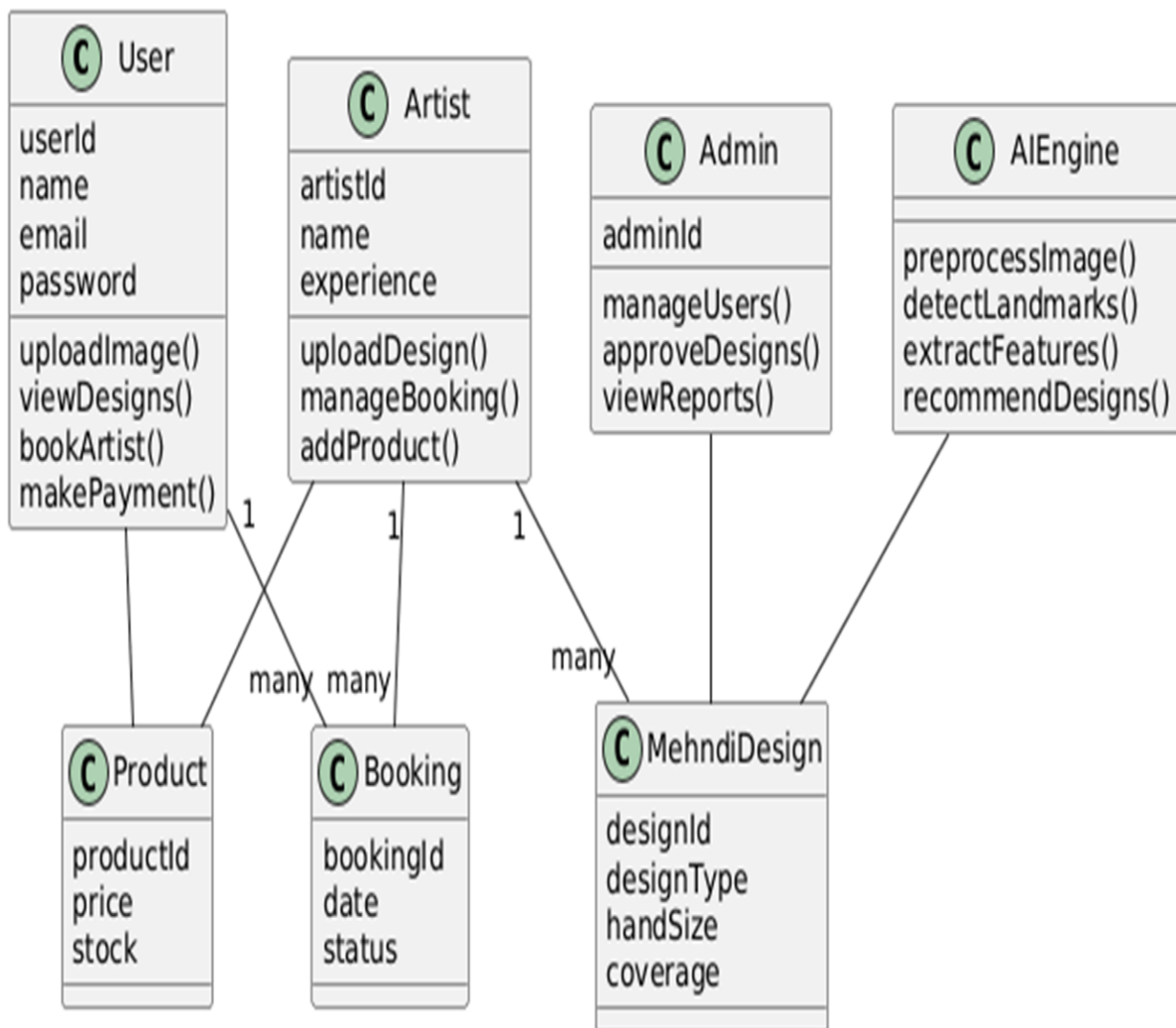


Figure 4.3.1 Class Diagram

User Class:

The User class represents customers who use the system to access mehndi-related services. These users can upload hand images, view recommended designs, book artists, and make payments.

Attributes:

- user Id : int
- name : string
- email : string

- password : string

Methods:

- upload Image() – Allows the user to upload a plain hand image for analysis.
- view Designs() – Displays the recommended mehndi designs.
- book Artist() – Enables the user to book a mehndi artist.
- make Payment() – Allows the user to complete payment for bookings or products.

Artist Class:

The Artist class represents mehndi artists who provide design services and sell henna products through the platform.

Attributes:

- artist Id : int
- name : string
- experience : string

Methods:

- upload Design() – Allows the artist to upload new mehndi designs.
- manage Booking() – Helps the artist manage customer booking requests.
- add Product() – Enables the artist to add henna products for sale.

Admin Class:

Admin class manages the overall functioning of the system. The administrator monitors users, approves designs, and views reports related to system activities.

Attributes:

- admin Id : int

Methods:

- manage Users() – Manages customer and artist accounts.
- approve Designs() – Approves uploaded mehndi designs before they are displayed.
- view Reports() – Allows the admin to monitor bookings, sales, and system reports.

AI Engine Class:

The AI Engine class is one of the most important classes in the system because it handles the artificial intelligence and image processing tasks required for recommendation.

Methods:

- preprocess Image() – Prepares the uploaded hand image for analysis.
- detect Landmarks() – Detects important hand landmarks such as fingers and palm points.
- extract Features() – Extracts features such as palm size and finger length.
- recommend Designs() – Recommends suitable mehndi designs based on extracted features.

Mehndi Design Class:

The Mehndi Design class represents the mehndi designs stored in the system database. These designs are categorized according to different characteristics for recommendation purposes.

Attributes:

- design Id : int
- design Type : string
- hand Size : string
- coverage : string

This class stores design-related information and is linked with the AI engine and artists.

Booking Class:

The Booking class manages the appointment details between users and mehndi artists.

Attributes:

- booking Id : int
- date : date
- status : string

This class stores booking information such as booking date and current status, helping both customers and artists track appointments.

Product Class:

The Product class represents henna products or related items sold by artists through the platform.

Attributes:

- product Id : int
- price : float
- stock : int

This class stores product details such as cost and available stock for sale.

The Class Diagram provides a structural overview of the AI-Based Mehndi Design Recommendation System by illustrating how different classes such as User, Artist, Admin, AI Engine, Mehndi Design, Booking, and Product interact with each other. These relationships define how information flows within the platform and how the various modules work together to support design recommendation, service booking, and product sales. Each class represents a specific entity within the system and contains attributes and methods that define its behavior.

For example, the User class handles uploading hand images, viewing recommended designs, booking artists, and making payments. The Artist class manages design uploads, bookings, and products, while the Admin class supervises the entire system. Supporting classes such as Booking, Product, and Mehndi Design store essential information required for transactions and recommendations. The AI Engine class plays a central role by processing hand images and generating personalized design suggestions. Overall, the Class Diagram acts as a blueprint for the system architecture and helps developers understand the internal structure of the platform, ensuring that all components work together efficiently to provide a reliable and intelligent mehndi recommendation system.

4.4 Activity Diagram

An Activity Diagram represents the workflow of a system by showing the sequence of actions performed by users and the system. It illustrates how different activities are executed step by step and how control flows from one activity to another. This diagram helps in understanding the overall operational process of the system.

In the AI-Based Mehndi Design Recommendation System, the activity diagram describes the complete flow of operations from the moment a user logs into the system until the final output such as booking or payment is completed. It mainly focuses on the customer interaction with the AI system and how the system processes hand images to generate personalized mehndi design recommendations.

- Start**

The process begins when the user opens the mehndi recommendation platform through a web interface or application.

- Login**

The user enters login credentials such as email and password to access the system.

- Upload Hand Image:** After logging in, the user uploads a plain hand image to the system for analysis.

- **Image Preprocessing:** The system processes the uploaded image by resizing, noise removal, and preparing it for further analysis.

- **Hand Landmark Detection:** The system detects important hand landmarks such as fingertips, joints, and palm regions using computer vision techniques.
- **Feature Extraction:** The system extracts key features such as palm width, finger length, and hand proportions from the detected landmarks.
- **Match Mehndi Designs:** Based on the extracted features, the system compares them with stored design attributes in the database.
- **Show Top 2–3 Designs:** The system recommends the most suitable mehndi designs (top 2–3 options) to the user.
- **User Selects Design?:** A decision point occurs where the user chooses whether to select a recommended design.
- **If Yes – Book Mehndi Artist:** If the user selects a design, they can proceed to book a mehndi artist through the system.
- **Make Payment:** After booking, the user completes a secure online payment for the selected service.
- **If No – Upload New Image:** If the user is not satisfied with the recommendations, they can upload a new hand image and repeat the process.
- **End:** The activity flow ends after the user completes the booking and payment process or exits the system.

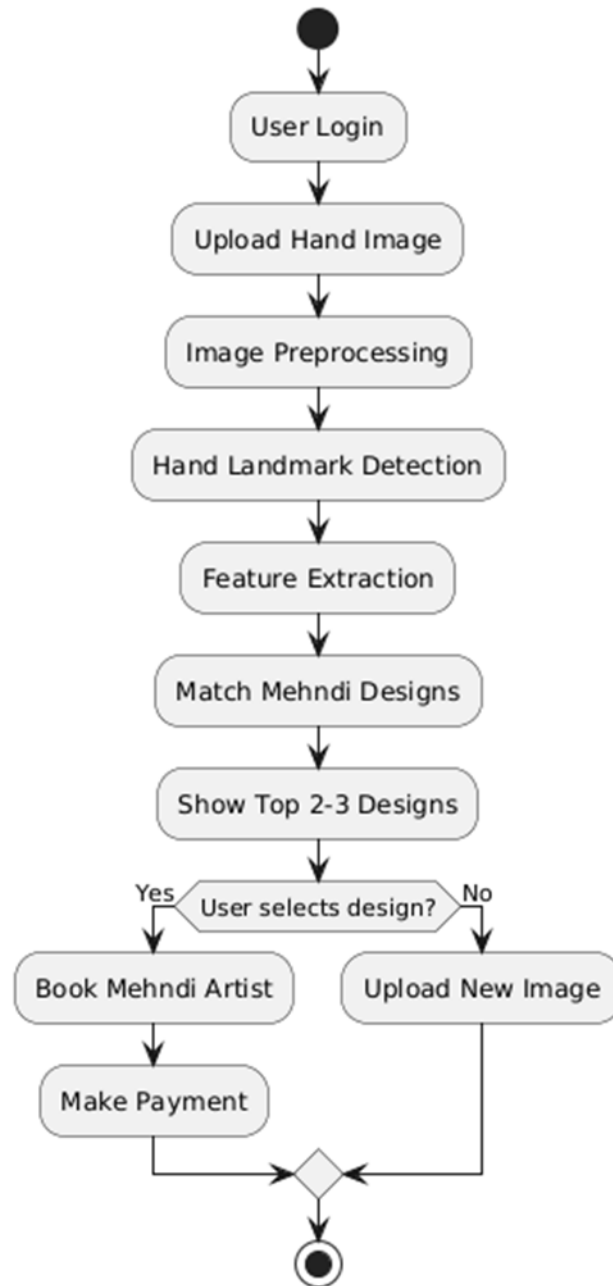


Figure 4.4.1 Activity Diagram

4.5 Sequence Diagram

The Sequence Diagram illustrates how different components of the system interact with each other over time to complete a particular task. It shows the order in which messages are exchanged between the user and the system modules. This diagram helps in understanding the communication flow between the web application, AI engine, design database, and artist during the mehndi recommendation and booking process.

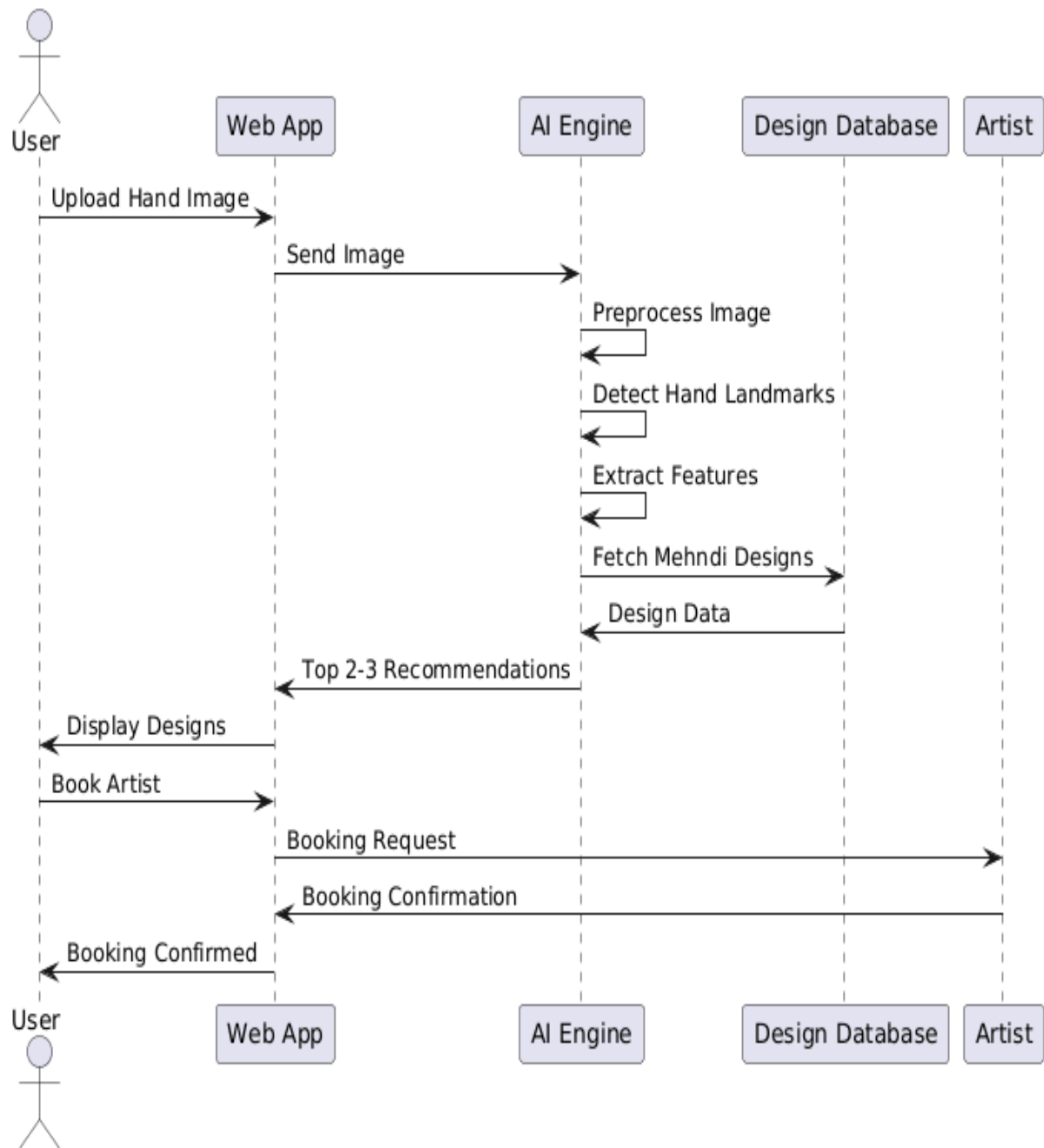


Figure 4.5.1 Sequence Diagram

In the AI-Based Mehndi Design Recommendation System, the main participants involved in the sequence diagram are:

- User
- Web App

- AI Engine
- Design Database
- Artist

The diagram explains how these actors and modules interact with the system to perform activities such as image upload, design recommendation, and booking confirmation.

• **Step 1: Upload Hand Image**

The user first uploads a plain hand image through the web application. The web app receives the image and forwards it to the AI engine for analysis.

• **Step 2: Image Processing**

The AI engine starts processing the uploaded image. It performs image preprocessing, detects hand landmarks, and extracts important hand features such as palm width, finger length, and hand proportions.

• **Step 3: Fetch Mehndi Designs**

After extracting the hand features, the AI engine sends a request to the design database to fetch mehndi designs that match the identified hand characteristics. The database returns the relevant design data to the AI engine.

• **Step 4: Generate Recommendations**

Using the extracted features and design data, the AI engine generates the top 2–3 most suitable mehndi design recommendations. These recommendations are sent back to the web application.

• **Step 5: Display Designs**

The web application displays the recommended mehndi designs to the user. The user can then browse the suggestions and choose a preferred design.

• **Step 6: Book Artist**

If the user decides to proceed, they send a booking request through the web app. The booking request is forwarded to the selected mehndi artist for confirmation.

- **Step 7: Booking Confirmation**

The artist receives the booking request and sends a confirmation response back to the web application. The web app then displays the booking confirmation message to the user.

- **Step 8: Final Output**

The system processes all requests from the user and returns the appropriate responses. These responses may include recommended designs, booking details, and confirmation messages.

The Sequence Diagram helps visualize the communication between the user, web application, AI engine, design database, and artist. It clearly represents how requests are sent, processed, and responded to within the system, providing a better understanding of the workflow of the AI-Based Mehndi Design Recommendation System.

CHAPTER 5

METHODOLOGY

The methodology describes the approach used to design, develop, and implement the AI-Based Mehndi Design Recommendation System. It explains how the system processes user inputs,

analyzes hand images using artificial intelligence techniques, and provides personalized mehndi design recommendations. The methodology also covers how different modules such as user management, design recommendation, booking, product sales, and payment processing work together to ensure smooth system functionality.

The proposed system follows a structured workflow that includes several stages such as user registration, authentication, hand image upload, image processing, feature extraction, design recommendation, booking management, and feedback handling. Each stage performs a specific function to ensure efficient interaction between customers, mehndi artists, and administrators. The system integrates computer vision and machine learning techniques to automate the process of selecting suitable mehndi designs based on hand geometry.

When users access the system, they first register and log in to the platform. After authentication, the system identifies the role of the user and provides access to the appropriate dashboard. Customers can upload plain hand images, receive AI-based design recommendations, browse different styles, book mehndi artists, and purchase henna products. Mehndi artists can upload designs, manage bookings, add products for sale, and track their earnings. Administrators oversee the entire system by managing users, approving designs and products, and monitoring system activities.

The system also includes database management for storing user information, mehndi design data, booking details, product records, and payment transactions. These stored data ensure that the system maintains proper records and allows efficient retrieval of information whenever required. By combining artificial intelligence, image processing, and web technologies, the methodology ensures accurate recommendations, efficient service management, and an improved user experience in the mehndi design selection process.

5.1 Module Descriptions

The AI-Based Mehndi Design Recommendation System is composed of several functional modules. Each module performs a specific task within the overall system workflow. These modules work together to manage user interactions, process hand images using artificial intelligence, provide personalized mehndi design recommendations, and support service booking, product sales, and system administration.

The system is organized in such a way that different users can access the platform according to their roles. Customers use the system to upload hand images, receive recommended designs, book artists, and purchase products. Mehndi artists manage design uploads, bookings, and product sales, while administrators control and monitor the overall system operations.

The system mainly consists of the following modules:

1. Admin Module
2. Artist Module
3. Customer Module

Each module is explained below.

5.1.1 Admin Module

The **Admin Module** controls and manages the overall operations of the AI-Based Mehndi Design Recommendation System. The administrator plays a crucial role in maintaining the system and ensuring that all activities are performed efficiently and securely. The admin has the highest level of access in the system and can monitor and manage all other modules.

One of the primary responsibilities of the administrator is user management. The admin can approve mehndi artist registrations, manage customer accounts, and maintain user records stored in the system database. This ensures that only authorized users can access the platform and perform their respective activities.

The administrator is also responsible for managing mehndi designs and products. This includes approving designs uploaded by artists, verifying product details, and ensuring that all content displayed on the platform meets quality standards. The admin can also monitor bookings, track transactions, and generate reports related to system usage, artist performance, and product sales. Another important feature of the admin module is system monitoring and feedback management. Customers and artists can submit feedback or complaints through the platform, and the administrator reviews these requests and provides appropriate responses. The admin also oversees online payment activities to ensure secure transactions and resolve any issues. Overall, the Admin Module ensures that the system operates smoothly while maintaining

reliability, security, and proper management of the AI-Based Mehndi Design Recommendation System.

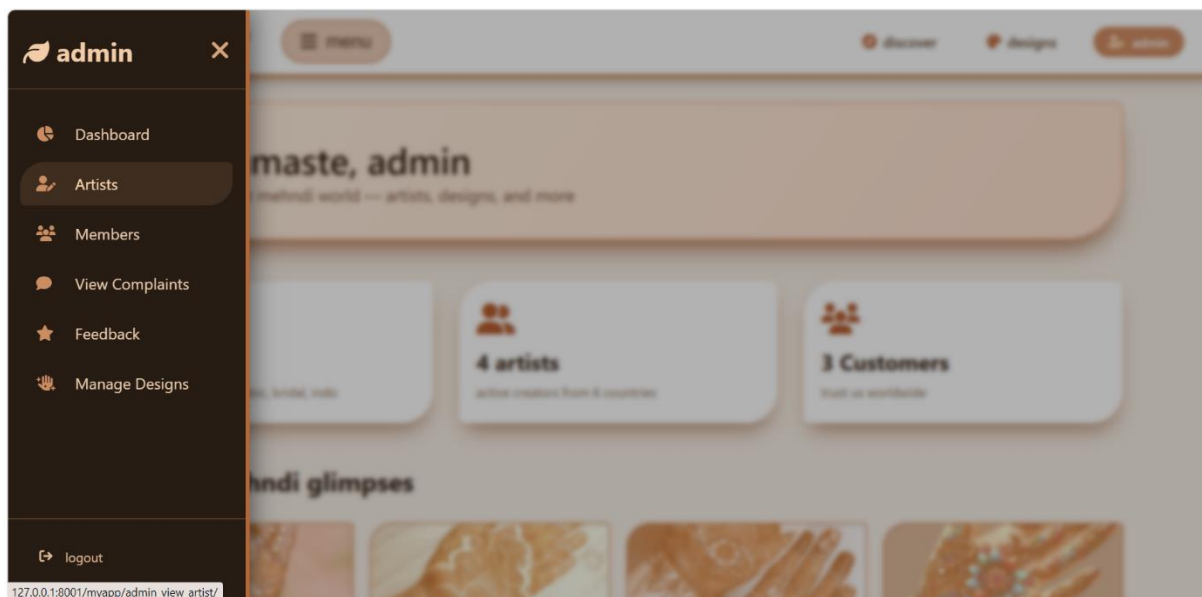


Figure 5.1.1 Admin Home Page

5.1.2 Artist Module

The **Artist Module** allows mehndi artists to manage their design portfolio, handle customer bookings, and sell henna-related products through the platform. Artists play a crucial role in providing creative services and interacting with customers. This module helps artists organize their designs, manage service requests, and grow their professional presence within the AI-Based Mehndi Design Recommendation System.

After logging into the system, artists can access their dashboard where they can upload mehndi designs such as Arabic, bridal, floral, or customized patterns. These designs are stored in the system database and are made available to customers after admin approval. Artists can also categorize designs based on style, hand size, and coverage, which helps the AI system provide more accurate recommendations to users.

Artists are also responsible for managing booking requests received from customers. They can view booking details, accept or reject requests, and schedule appointments accordingly. This helps artists efficiently manage multiple clients and maintain a structured workflow. The

system also allows artists to track booking history and manage their availability, ensuring smooth service delivery.

Another important function of the Artist Module is product management. Artists can add henna products such as cones, aftercare items, and accessories for sale. They can update product details, pricing, and stock availability. The system records product orders and enables artists to track their sales and earnings. This integration of service booking and product sales enhances business opportunities for artists.

The module also enables artists to view customer feedback and ratings, helping them improve their services based on user experience. Artists can respond to customer queries and maintain communication through the platform. Overall, the Artist Module supports efficient management of designs, bookings, and product sales while helping artists expand their reach and provide better services to customers.

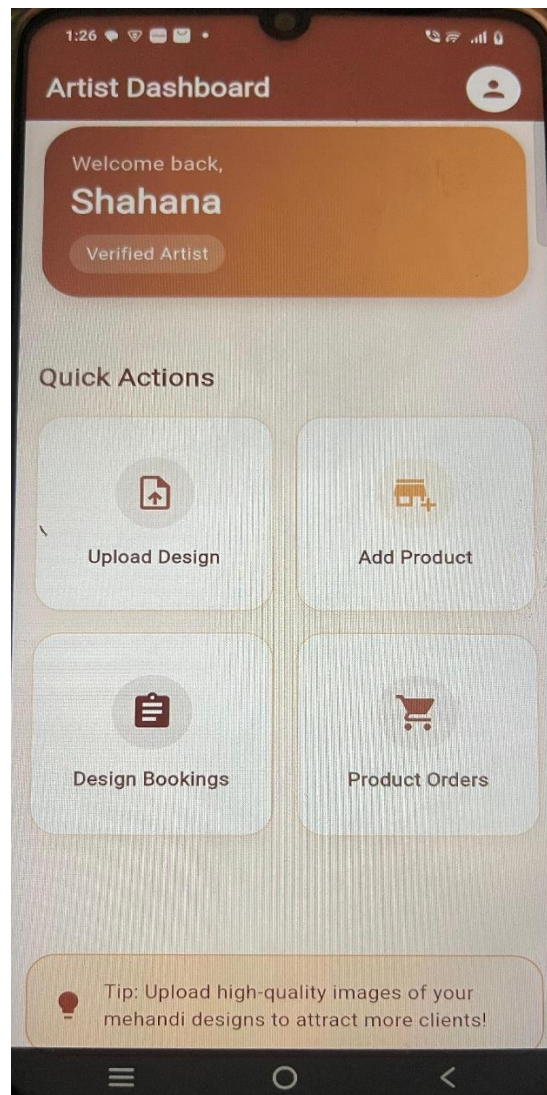


Figure 5.1.2 Artist Home Page

5.1.3 User(customer) Module

The **Customer Module** is designed to provide users with easy access to mehndi design services and personalized recommendations through the AI-Based Mehndi Design Recommendation System. This module acts as the primary interface through which customers interact with the system and access various features such as design recommendation, artist booking, and product purchase. Customers can register in the system and log in using their credentials to access the available services provided by the platform.

Once logged in, customers can upload a plain hand image to the system. The uploaded image is processed using artificial intelligence and computer vision techniques, including image

preprocessing, hand landmark detection, and feature extraction. Based on these extracted features, the system generates personalized mehndi design recommendations that best suit the user's hand structure. Customers can view these recommended designs, explore different styles, and select their preferred options.

Customers can also browse available mehndi artists and view their design portfolios, experience, and ratings. This helps users choose suitable artists for their requirements. The system allows customers to send booking requests, schedule appointments, and manage their bookings efficiently. Customers can track booking status and receive notifications regarding confirmations or updates from artists.

Another important function of this module is product purchasing. Customers can browse henna products listed by artists, view product details such as price and availability, and place orders through the platform. The system supports secure online payment for both service bookings and product purchases, ensuring a smooth and reliable transaction process.

The module also includes a feedback and rating system. Customers can provide feedback on designs, services, and products, helping improve system quality and artist performance. They can also submit complaints if any issues arise, and track the status of their requests.

In addition, the Customer Module allows users to view their activity history, including uploaded images, recommended designs, booking records, and product orders. This helps customers manage their interactions with the system and make informed decisions in the future. Overall, the Customer Module creates a convenient and user-friendly environment where customers can easily access mehndi services, receive personalized recommendations, and interact efficiently with artists through a centralized digital platform.

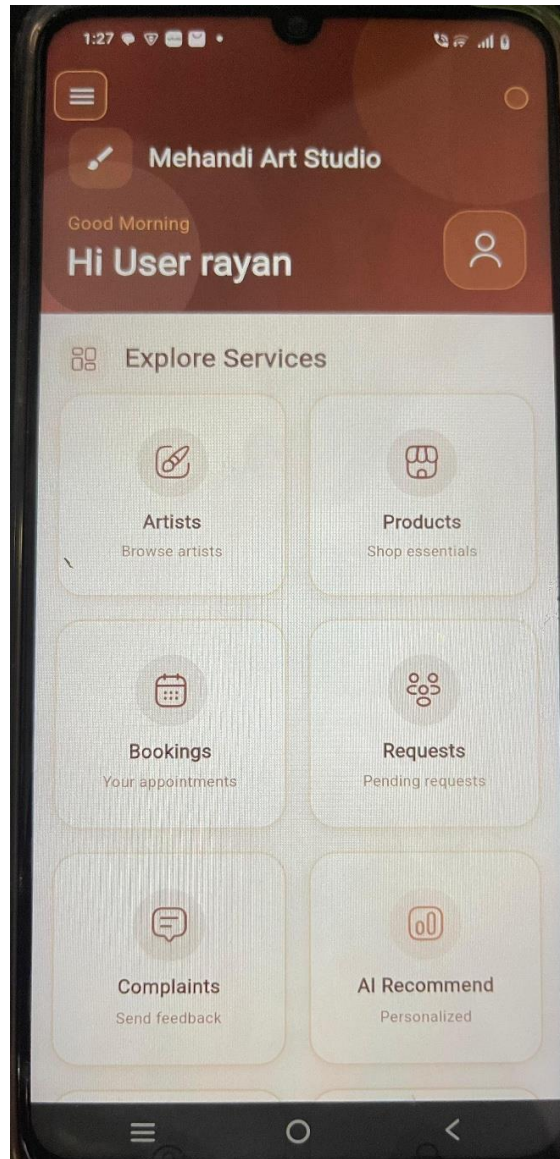


Figure 5.1.3 User(customer) Page

5.2 Algorithms

The AI-Based Mehndi Design Recommendation System uses artificial intelligence and computer vision algorithms to analyze hand images and recommend suitable mehndi designs based on hand geometry and similarity matching. The system integrates landmark detection and distance-based recommendation techniques to process user input and generate personalized results.

The primary algorithms used in the system include:

- Media Pipe Hand Landmark Detection Algorithm
- Euclidean Distance-Based Recommendation Algorithm

These algorithms work together to extract hand features from the uploaded image and compare them with stored mehndi design features to provide accurate recommendations.

• **Media Pipe Hand Landmark Detection Algorithm**

The Media Pipe Hand Landmark Detection Algorithm is used to detect key points on the hand such as fingertips, joints, palm center, and wrist. This algorithm is implemented using the Media Pipe framework along with OpenCV for image processing. It helps the system understand the structure of the user's hand by identifying 21 important landmarks.

In the system, the uploaded hand image is first read using OpenCV and converted into RGB format. The Media Pipe model processes the image and detects hand landmarks. Using these landmarks, distances between key points are calculated to extract geometric features such as palm width, palm length, and finger lengths.

Steps of Media Pipe Hand Landmark Detection Algorithm

1. Capture the uploaded hand image using OpenCV.
2. Convert the image from BGR to RGB format.
3. Apply MediaPipe Hands model to detect hand landmarks.
4. Identify 21 key points such as fingertips, joints, and palm.
5. Calculate distances between landmarks using Euclidean distance formula.
6. Extract features such as palm width, palm length, and finger lengths.
7. Store the extracted features for further processing.

This algorithm ensures accurate detection of hand structure using Media Pipe and helps in extracting meaningful features. These features are essential for generating personalized mehndi design recommendations based on hand shape and proportions. The integration of OpenCV and Media Pipe ensures reliable and scalable performance.

• **Euclidean Distance-Based Recommendation Algorithm**

The Euclidean Distance-Based Recommendation Algorithm is used to compare the user's hand features with the stored mehndi design features in the database. This algorithm calculates the similarity between feature vectors and identifies the closest matching designs.

In the system, both user features and design features are converted into numerical vectors using NumPy arrays. The Euclidean distance formula is applied to measure how similar the user's hand is to each stored design. Designs with the smallest distance values are considered the best matches. The system then sorts the results, applies randomization for similar matches, and returns the top recommended designs.

Steps of Euclidean Distance-Based Recommendation Algorithm

1. Convert user hand features into a numerical vector.
2. Retrieve mehndi design features from the database.
3. Convert each design's features into a vector.
4. Calculate Euclidean distance between user vector and design vector using:
$$\text{distance} = \sqrt{\sum (\text{user_feature} - \text{design_feature})^2}$$
5. Store distance values for each design.
6. Sort designs based on minimum distance (closest match).
7. Shuffle similar results to add variation.
8. Return the top 5 recommended mehndi designs to the user.

This algorithm ensures accurate and efficient matching of hand features with suitable mehndi designs, reducing manual effort and improving user satisfaction by providing personalized recommendations.

5.3 System Architecture

The system architecture of the AI-Based Mehndi Design Recommendation System describes the overall structural design and workflow of the proposed system. The architecture explains how different components of the platform interact with each other to support design recommendation, service booking, product sales, and communication between users. The system is designed to provide a centralized platform where administrators, mehndi artists, and customers can perform their respective activities efficiently.

The architecture follows a structured framework where the AI-based system acts as the central processing unit connecting the main actors of the system: Admin, Artist, and Customer. These actors interact with the platform through a web-based interface. All requests generated by users are processed by the system, and the corresponding information is stored in the system database. This architecture ensures smooth data flow between users and the database while maintaining system reliability, scalability, and security.

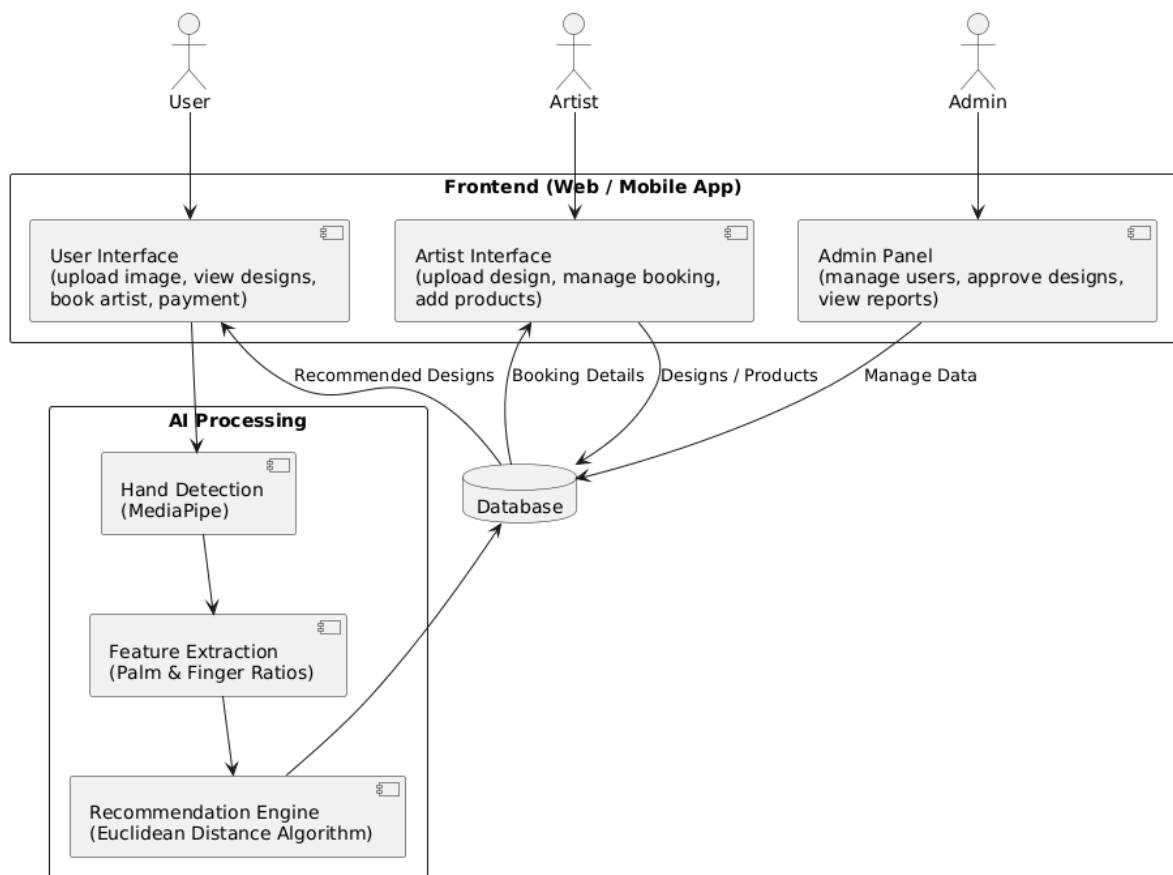


Figure 5.3.1 Architecture

The workflow begins when a user accesses the platform through the web interface. The user first performs authentication by entering login credentials. The system verifies the credentials using the login module stored in the database. After successful authentication, the user is redirected to the appropriate dashboard based on their role, such as administrator, artist, or customer.

The Admin module manages the overall functioning of the system. Administrators are responsible for controlling system operations such as managing users, approving artist registrations, verifying mehndi designs and products, monitoring bookings and payments, and reviewing system reports. The admin also handles feedback submitted by customers and artists, ensuring smooth operation of the platform. All administrative data is stored in the database for tracking and management purposes.

The Artist module allows mehndi artists to manage their services and design portfolios. Artists can upload mehndi designs, categorize them based on style and coverage, manage booking requests, and add henna products for sale. They can also track their earnings from both bookings and product sales. The uploaded designs and product details are stored in the database, making them accessible to customers through the system interface.

The Customer module provides access to personalized mehndi services. Customers can log in to the system, upload hand images, and receive AI-based design recommendations. The system processes the uploaded image using image preprocessing, hand landmark detection, and feature extraction techniques, and matches the extracted features with stored design data. Customers can view recommended designs, book artists, purchase products, and make secure online payments. They can also provide feedback and track their activity history.

All system data is maintained within a centralized database module. The database stores records such as login details, artist and customer information, mehndi designs, product details, booking information, AI recommendations, hand images, and feedback. This structured storage ensures efficient retrieval and updating of data whenever required.

The system also includes AI processing and reporting capabilities. The AI engine performs image analysis and recommendation generation, while the system tracks bookings, product sales, and user activities. These features help administrators and artists analyze system performance, customer preferences, and business growth. Overall, the system architecture provides a reliable, scalable, and intelligent platform for automating mehndi design recommendation and service management

CHAPTER 6

IMPLEMENTATION DETAILS

The implementation phase involves converting the system design and methodology into a fully functional software application. In this project, the AI-Based Mehndi Design Recommendation System was implemented using web development technologies and artificial intelligence frameworks. The system integrates database management, web interfaces, and computer vision techniques to create a complete platform for mehndi design recommendation, service booking, and product sales.

The implementation focuses on developing a structured platform that allows different users such as administrators, mehndi artists, and customers to interact with the system through a web-based interface. The system handles multiple functions including user authentication, hand image processing, design recommendation, booking management, product sales, and communication between users. It also integrates image processing and machine learning techniques to analyze hand features and generate personalized mehndi design suggestions.

The platform was developed using the Python programming language along with the Django web framework for backend development. Frontend technologies such as HTML, CSS, and JavaScript were used to create an interactive and user-friendly interface. The backend system processes user requests, performs AI-based computations, and communicates with the database to store and retrieve information. The database stores important data such as user accounts, artist details, mehndi designs, booking records, product details, payment transactions, and feedback messages.

In addition, the system integrates computer vision libraries such as OpenCV and MediaPipe for hand landmark detection and feature extraction. A Euclidean Distance-based algorithm is used to compare hand features with stored design attributes and generate accurate recommendations. These technologies enable the system to analyze hand images effectively and provide personalized results to users.

The implementation process includes system setup, module development, database integration, and testing of different system components. Each module such as admin, artist, customer, AI processing, booking, and payment was developed and tested individually before integrating them into the complete system. This chapter explains the technologies used in the project, the

development environment, the implementation steps of different modules, and the testing procedures used to verify the functionality and performance of the AI-Based Mehndi Design Recommendation System.

6.1 Implementation Procedures

The implementation of the AI-Based Mehndi Design Recommendation System was carried out in a systematic and structured manner to ensure that all components of the system function efficiently and are properly integrated. The implementation phase focused on transforming the system design into a working application by developing individual modules, integrating artificial intelligence techniques, and connecting the system with a centralized database.

The system was developed using the Python programming language due to its flexibility, ease of use, and strong support for artificial intelligence and web development. The Django framework was used for backend development, enabling efficient handling of user requests, session management, and database interaction. The frontend interface was developed using HTML, CSS, and JavaScript, providing a responsive and user-friendly experience for customers, artists, and administrators.

The implementation process involved multiple stages including environment setup, dataset preparation, preprocessing of images, feature extraction, recommendation model development, and user interface integration. Each module was developed independently and tested before being integrated into the final system. This modular approach ensured better maintainability, scalability, and error handling throughout the system.

6.1.1 Development Environment

The development environment plays a crucial role in ensuring smooth implementation and execution of the system. The AI-Based Mehndi Design Recommendation System was developed using a combination of software tools, frameworks, and libraries that support artificial intelligence, image processing, and web development.

Programming Language

Python was selected as the primary programming language due to its extensive library support and ease of integration with AI and web technologies. Python enables efficient handling of numerical computations, image processing tasks, and backend logic implementation.

Framework

The system was implemented using the Django framework, which provides built-in features such as authentication, database management, and URL routing. Django follows the Model-View-Template (MVT) architecture, making it easier to organize the system into logical components.

Frontend Technologies

The user interface was developed using HTML, CSS, and JavaScript. These technologies ensure that the platform is interactive, visually appealing, and accessible across different devices.

Libraries Used

- NumPy – Used for numerical operations and vector calculations in recommendation algorithms.
- OpenCV – Used for image reading, processing, and format conversion.
- Media Pipe – Used for detecting hand landmarks and extracting hand features.
- Math Library – Used for calculating Euclidean distances between landmark points.
- Django ORM – Used for efficient database operations and data management.

This combination of tools ensures that the system is robust, scalable, and capable of handling real-time image processing and user interactions.

6.1.2 Dataset Collection

The dataset plays an important role in the performance of the recommendation system. In this project, the dataset consists of mehndi designs along with associated feature values that

represent different hand structures. The dataset is stored in the system database and is used for matching user input with appropriate designs.

The dataset includes:

- Mehndi design images categorized into styles such as Arabic, bridal, floral, and simple designs
- Feature values such as palm width ratio, palm length ratio, and finger lengths
- Design attributes like coverage level and suitability for different hand sizes

These features are either manually assigned or derived from sample hand images. The dataset is continuously updated as artists upload new designs to the platform. This dynamic dataset helps improve recommendation accuracy over time.

The quality of the dataset directly affects the performance of the system. Therefore, proper categorization and feature representation are maintained to ensure accurate matching between user hand features and mehndi designs.

6.1.3 Data Preprocessing Implementation

Data preprocessing is an essential step before applying computer vision algorithms. Raw images uploaded by users may contain noise, varying resolutions, and background distractions, which can affect the accuracy of feature extraction.

The preprocessing stage ensures that the image is cleaned and prepared for analysis. The following steps were implemented:

Image Resizing: All uploaded images are resized to a standard dimension to ensure consistency during processing. This helps improve computational efficiency and ensures compatibility with the AI model.

Color Conversion: Images are converted from BGR format to RGB format using OpenCV. This is necessary because Media Pipe processes images in RGB format.

Noise Reduction: Basic filtering techniques are applied to remove noise and improve image clarity. This helps in better detection of hand landmarks.

Hand Region Detection: The system focuses only on the hand region by detecting the presence of a hand in the image. If no hand is detected, the system prompts the user to upload a valid image.

6.1.4 Feature Extraction Implementation

Feature extraction is a critical stage in the system where meaningful information is derived from the processed hand image. The system uses Media Pipe Hand Landmark Detection to identify 21 key points on the hand.

Using these landmarks, geometric features are calculated using the Euclidean distance formula. These features represent the structure of the hand and are used for design recommendation.

Extracted Features Include:

- Palm width (distance between landmarks 5 and 17)
- Palm length (distance between wrist and middle finger base)
- Finger lengths (index, middle, ring, pinky , thumb)

The feature extraction process converts visual data into numerical values that can be used for comparison. These numerical features are stored temporarily and passed to the recommendation module.

Accurate feature extraction ensures that the system understands the user's hand shape correctly, which directly impacts the quality of recommendations.

6.1.5 Recommendation Algorithm Implementation

The recommendation system is implemented using a Euclidean Distance-Based Similarity Algorithm, which compares the user's hand features with stored design features. The Euclidean distance-based recommendation algorithm provides a simple and effective way to match user

First, the user's features are converted into a vector using NumPy. Similarly, each mehndi design in the database is represented as a feature vector. The system calculates the Euclidean distance between these vectors to measure similarity.

Designs with the smallest distance values are considered the best matches. The system sorts these results and selects the top designs. To avoid repetitive results, a random shuffle is applied to similar matches before returning the final recommendations.

This algorithm is efficient, easy to implement, and provides accurate results for feature-based matching. It ensures that users receive personalized design suggestions based on their unique hand structure.

6.1.6 Web Interface Implementation

The web interface provides the interaction layer between users and the system. It was designed to be simple, responsive, and easy to use.

Login Section: Users can register and log in using their credentials. Authentication is handled by Django's built-in system.

Dashboard Section: Displays different functionalities based on user roles such as customer, artist, or admin.

Image Upload Section: Allows users to upload hand images for AI-based analysis.

Recommendation Section: Displays the top recommended mehndi designs with images and details.

Booking Section: Users can select artists and schedule appointments. Booking details are stored in the database.

Product & Payment Section: Customers can purchase henna products and complete secure payments through the platform.

Feedback Section: Users can provide feedback and track responses from the admin.

The interface ensures smooth navigation and enhances user experience by providing all services in a centralized platform.

6.2 Testing Methods and Results

Testing is an important phase in the implementation process because it ensures that the system functions correctly and performs as expected. In the AI-Based Mehndi Design Recommendation System, testing was carried out to verify the performance of the image processing algorithms, recommendation system, and overall functionality of all modules such as admin, artist, and customer. Several testing techniques were used to evaluate the performance of the system, including dataset testing, algorithm evaluation, and overall system functionality testing.

6.2.1 Dataset Testing

The dataset used for the recommendation system consists of mehndi designs along with associated hand feature values. The dataset was divided into two parts to evaluate system performance effectively:

- **Training Dataset**

Used to store and define mehndi design features and attributes in the database.

- **Testing Dataset**

Used to evaluate the performance of the recommendation system using new and unseen hand images.

A common dataset division method known as the 80–20 split was used:

- 80% of the dataset was used to define and store design feature values.
- 20% of the dataset was used for testing the system with new hand images.

This approach ensures that the system is evaluated using unseen inputs, providing a more realistic measure of recommendation accuracy.

6.2.2 Model Evaluation Metrics

Several evaluation metrics were used to measure the performance of the recommendation system and feature extraction process.

Accuracy

Accuracy measures how correctly the system recommends suitable mehndi designs based on user hand features.

$$\text{Accuracy} = (\text{Correct Recommendations} / \text{Total Recommendations}) \times 100$$

Precision

Precision measures how many of the recommended designs are actually relevant to the user's hand features.

$$\text{Precision} = \text{True Positives} / (\text{True Positives} + \text{False Positives})$$

Recall

Recall measures how many suitable designs present in the dataset were successfully recommended by the system.

$$\text{Recall} = \text{True Positives} / (\text{True Positives} + \text{False Negatives})$$

F1Score

The F1 Score provides a balanced measurement by combining precision and recall.

$$\text{F1 Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

These metrics help evaluate how effectively the system matches user hand features with appropriate mehndi designs.

6.2.3 Confusion Matrix

A confusion matrix was used to evaluate the performance of the recommendation system. It provides a detailed comparison between actual suitable designs and recommended designs.

The confusion matrix consists of four components:

- True Positive (TP) – Correct design recommended by the system
- True Negative (TN) – Incorrect design correctly not recommended
- False Positive (FP) – Incorrect design recommended

- False Negative (FN) – Suitable design not recommended

The confusion matrix helps visualize the accuracy and reliability of the recommendation algorithm and highlights areas for improvement.

6.2.4 System Testing

In addition to evaluating the performance of the algorithms, the entire system was tested to ensure that all modules function correctly and work together without errors. Different types of system testing were carried out to verify reliability and functionality. The login and authentication module was tested to ensure that users can successfully register and log in using valid credentials and are redirected to the correct dashboard based on their role (admin, artist, or customer).

The image processing module was tested to confirm that uploaded hand images are correctly processed, and hand landmarks are accurately detected using Media Pipe. The feature extraction process was also verified to ensure correct calculation of palm width, palm length, and finger lengths. The recommendation module was tested to ensure that the system correctly compares user features with stored design features using the Euclidean distance algorithm and returns the most relevant designs.

The booking and product modules were tested to verify that customers can successfully book artists, purchase products, and complete secure online payments without errors. Additionally, database testing was performed to ensure that all system data such as user details, designs, bookings, product information, and feedback are correctly stored and retrieved whenever required.

6.2.5 Testing Results

The testing process showed that the AI-Based Mehndi Design Recommendation System operates correctly and that all modules function as expected. The user authentication system successfully verified login credentials and redirected users to their respective dashboards without errors.

The image processing module accurately detected hand landmarks and extracted relevant features from uploaded images. The recommendation system successfully matched user features with stored design data and generated appropriate mehndi design suggestions.

The booking and product modules allowed customers to book artists and purchase products smoothly, and the payment system processed transactions securely. Database testing confirmed that all data such as user information, designs, bookings, and feedback were stored and retrieved accurately.

Overall, the testing results indicate that the system performs reliably and provides an efficient and user-friendly platform for mehndi design recommendation, service booking, and product sales.

CHAPTER 7

ADVANTAGES AND DISADVANTAGES

The AI-Based Mehndi Design Recommendation System was designed to modernize the traditional mehndi selection process by integrating web technologies and artificial intelligence techniques. The system uses computer vision algorithms and feature-based recommendation methods to analyze hand images and provide personalized mehndi design suggestions. It also

integrates additional features such as artist booking, product sales, and online payment to create a complete digital platform.

While the system provides several advantages in terms of automation, personalization, and ease of use, it also has certain limitations due to technological constraints, dependency on image quality, and the simplicity of the recommendation algorithm. This chapter discusses the advantages and limitations of the proposed system to provide a balanced evaluation of its performance and practical usability.

7.1 Advantages

The AI-Based Mehndi Design Recommendation System offers several advantages in terms of automation, personalization, and efficient service management. The system provides a centralized platform for recommending mehndi designs, managing bookings, and facilitating product sales while integrating artificial intelligence for accurate decision-making.

7.1.1 Automated Design Recommendation System

One of the major advantages of the system is its ability to automatically recommend suitable mehndi designs based on the user's hand structure. The platform uses computer vision techniques and feature extraction methods to analyze hand images and generate personalized design suggestions.

This automated process reduces manual effort, saves time, and helps users quickly select designs without browsing through large collections.

7.1.2 AI-Based Hand Image Analysis

The system uses advanced technologies such as MediaPipe and image processing algorithms to detect hand landmarks and extract features like palm width and finger length.

This intelligent analysis improves the accuracy of recommendations and ensures that the suggested designs match the user's hand shape, providing a more customized experience.

7.1.3 Efficient Booking Management

The platform provides an integrated booking system that allows customers to easily connect with mehndi artists and schedule appointments.

This reduces communication gaps between users and artists and ensures efficient service management. Artists can manage bookings, track requests, and organize their schedules effectively.

7.1.4 Integrated Product Sales and Online Payment

The system includes features for selling henna products and supports secure online payment methods. Customers can purchase products and complete transactions within the same platform.

This integration creates a complete digital ecosystem for both services and product sales, improving convenience and business opportunities for artists.

7.1.5 User-Friendly Interface

The system provides a simple and interactive web interface that allows users to easily navigate through different features such as image upload, design recommendation, booking, and product purchase.

The intuitive interface ensures that users can operate the system without requiring advanced technical knowledge.

7.1.6 Modular System Architecture

The system follows a modular architecture where different modules such as Admin, Artist, Customer, AI Processing, Booking, and Payment operate independently.

This modular design allows easy modification and scalability. New features such as advanced AI models or mobile applications can be added without affecting existing components.

7.1.7 Centralized Data Management

The system maintains a centralized database that stores important information such as user details, mehndi designs, booking records, product data, payment transactions, and feedback.

This ensures efficient data storage, easy retrieval, and better system management. It also helps administrators monitor system performance and maintain data consistency.

7.2 Disadvantages

Despite the many advantages offered by the AI-Based Mehndi Design Recommendation System, the system also has certain limitations. These limitations arise mainly due to technological constraints, dependency on image quality, and the simplicity of feature-based recommendation methods. Understanding these limitations helps identify areas where the system can be improved in future developments.

7.2.1 Dependence on Dataset Quality

The accuracy of the recommendation system depends greatly on the quality and diversity of the mehndi design dataset. If the dataset contains limited design styles or lacks variation in hand features, the system may not provide accurate recommendations. For example, if the database does not include sufficient designs for different hand shapes or sizes, the system may fail to match user features correctly, resulting in less relevant suggestions.

7.2.2 Limitations in Recommendation Accuracy

The system uses a Euclidean distance-based algorithm for matching hand features with design attributes. While this method is simple and efficient, it may not always capture complex relationships between hand structure and design aesthetics.

In some cases, the system may recommend designs that are mathematically similar but not visually or stylistically optimal for the user.

7.2.3 Dependence on Image Quality

The system relies on uploaded hand images for feature extraction. If the image is blurred, poorly lit, or partially visible, the system may fail to detect hand landmarks accurately.

Such issues can lead to incorrect feature extraction and affect the quality of recommendations provided by the system.

7.2.4 Computational Requirements

Although the system does not use heavy deep learning models, image processing and feature extraction still require computational resources. Processing multiple images simultaneously or handling large datasets may increase system load.

For large-scale deployment, additional hardware resources or optimized algorithms may be required to maintain performance.

7.2.5 Internet Dependency

The system operates as a web-based platform, which requires a stable internet connection for uploading images, processing data, booking services, and completing payments.

If the internet connection is slow or unstable, users may experience delays or interruptions while using the system.

7.2.6 Limited Personalization Factors

The system primarily focuses on hand geometry for recommendation and does not fully consider other important factors such as user preferences, occasion (wedding, festival), cultural styles, or skin tone compatibility.

This limitation may reduce the personalization level of the recommendations in certain scenarios.

7.2.7 Security and Privacy Concerns

Since the system stores user data such as images, booking details, and payment information, there may be concerns related to data security and privacy.

It is important to implement proper security measures to protect user data and ensure safe handling of sensitive information.

Summary:

The AI-Based Mehndi Design Recommendation System provides an effective solution for automating mehndi design selection and integrating service booking and product sales into a single platform. The system improves user experience through personalization, automation, and digital convenience.

However, certain challenges remain, such as dependency on dataset quality, limitations of the recommendation algorithm, and the need for good image quality and stable internet connectivity. Additionally, factors such as user preferences and cultural variations are not fully considered in the current system.

Overall, the system demonstrates the potential of applying artificial intelligence and computer vision in creative domains. Understanding both the advantages and limitations helps developers improve the system in the future by incorporating advanced machine learning models, enhanced datasets, and additional personalization features.

CHAPTER 8

RESULTS AND DISCUSSION

The Results and Discussion chapter presents the outcomes obtained from the implementation of the AI-Based Mehndi Design Recommendation System. This chapter evaluates how effectively the system performs hand image analysis, generates personalized mehndi design recommendations, and manages additional functionalities such as artist booking and product sales through an integrated platform.

The main objective of this project is to develop a system that automates the process of selecting suitable mehndi designs based on hand features while providing a complete digital solution for service booking and product purchasing. The system integrates web technologies with artificial intelligence techniques such as MediaPipe hand landmark detection and a Euclidean distance-based recommendation algorithm to analyze hand images and generate appropriate design suggestions.

The implemented system processes user interactions through several modules including user authentication, image upload, feature extraction, design recommendation, booking management, and payment processing. When a user uploads a hand image, the system preprocesses the image and detects hand landmarks using MediaPipe. The extracted features such as palm width, palm length, and finger lengths are then used by the recommendation algorithm to compare with stored design data and identify the most suitable mehndi designs.

The experimental results demonstrate that the system is capable of accurately detecting hand landmarks and extracting relevant features from uploaded images. The recommendation system successfully matches user hand features with stored design attributes and provides personalized design suggestions. The system is able to recommend the top matching designs efficiently, reducing the time required for manual selection.

The results also show that the integration of booking and product modules enhances the overall functionality of the platform. Users can seamlessly book mehndi artists, purchase henna products, and complete online payments within the same system. The database effectively stores and retrieves user data, design information, booking records, and transaction details, ensuring smooth system operation.

Overall, the system demonstrates that combining artificial intelligence, image processing, and web technologies can significantly improve the mehndi design selection process. The platform provides a reliable, efficient, and user-friendly solution that benefits both customers and artists. The results confirm that the proposed system successfully achieves its objectives and offers a practical application of AI in the creative and cultural domain.

8.1 System Output Overview

When a user interacts with the system, it processes the request and generates outputs based on the selected module. These outputs help users understand the results of their actions within the platform.

The main outputs generated by the system include:

- Successful user authentication and dashboard access
- Display of AI-recommended mehndi designs based on uploaded hand image
- Artist details, booking confirmation, and appointment information
- Product listings, order details, and online payment confirmation
- System notifications, feedback responses, and activity updates

The system interface presents these outputs through an interactive dashboard, allowing users to easily access and manage system features.

8.2 Monitoring Results

The system analyzes uploaded hand images using computer vision techniques to extract hand features and generate design recommendations.

During processing, the system performs image preprocessing, hand landmark detection, and feature extraction. If the hand image is clear and properly captured, the system successfully detects landmarks and generates accurate design recommendations. This is classified as a normal processing condition.

However, if the image quality is poor due to low lighting, blur, or incomplete hand visibility, the system may not detect landmarks accurately. In such cases, the result may be classified as low visibility or unclear image, and the system may prompt the user to upload a better image.

The testing results show that the system can effectively process hand images and generate recommendations in real time, demonstrating the efficiency of the AI-based approach.

8.3 User Interaction Results

The system supports interaction between customers, mehndi artists, and administrators. Customers can upload images, view recommended designs, book artists, and purchase products through the dashboard.

Artists can upload designs, manage bookings, and track product sales, while administrators can manage users, approve content, and monitor system activities. The system successfully retrieves and displays all required information from the database.

These interactions demonstrate that the platform effectively supports communication, service management, and user engagement within the system

8.4 Recommendation Score

The system generates recommendation results based on the similarity between user hand features and stored design features.

The recommendation score depends on several factors, including:

- Similarity between user hand features and design data
- Euclidean distance between feature vectors
- Accuracy of hand landmark detection
- Consistency of extracted measurements

Lower distance values indicate higher similarity and better recommendations, while higher values indicate less suitable matches. The system ranks designs based on these scores and presents the top matching designs to the user.

8.5 Feature Detection Results

The system performs feature detection to identify key hand landmarks in the uploaded image. The Media Pipe algorithm detects points such as fingertips, joints, palm center, and wrist position.

These landmarks are used to calculate features such as palm width, palm length, and finger lengths. The detected features are converted into numerical values and used in the recommendation process.

The system accurately detects hand structures in most test cases, confirming the effectiveness of the feature extraction process.

8.6 Visualization, Analytics, System Performance Evaluation, and Discussion

Visualization and analytics improve the usability and transparency of the system by presenting results in an understandable format. The system displays recommended designs, booking details, and product information through a user-friendly interface.

The system performance was evaluated based on factors such as recommendation accuracy, response time, and system reliability. The AI-based feature extraction and recommendation algorithm demonstrated effective performance in generating relevant design suggestions.

The system processes images within a short time interval, enabling near real-time recommendations. This ensures a smooth and efficient user experience.

Overall, the results show that the AI-Based Mehndi Design Recommendation System successfully integrates image processing, recommendation algorithms, and web-based functionalities. The system provides a reliable platform for mehndi design selection, artist booking, and product sales. The discussion highlights that future improvements such as advanced AI models, larger datasets, and enhanced personalization can further improve system performance and user satisfaction.

CHAPTER 9

CONCLUSION

9.1 Summary

The rapid growth of digital technologies and online platforms has created new opportunities for automating traditional processes in various domains, including art and design. In the field of mehndi design selection, users often face difficulty in choosing suitable designs based on their hand structure and personal preferences. Traditional methods require manual searching and expert suggestions, which can be time-consuming and less efficient. This project aimed to develop an intelligent system that automates mehndi design recommendation while also providing additional services such as artist booking and product purchasing.

The proposed AI-Based Mehndi Design Recommendation System integrates web technologies with artificial intelligence techniques to create a comprehensive and user-friendly platform. The system allows administrators, mehndi artists, and customers to interact through a centralized platform where designs, bookings, and product sales can be efficiently managed. Customers can upload hand images, receive personalized design recommendations, and access various services through a single interface.

The platform uses computer vision techniques such as MediaPipe hand landmark detection to analyze hand images and extract important features like palm width, palm length, and finger lengths. These extracted features are then compared with stored design attributes using a Euclidean distance-based algorithm to generate suitable mehndi design recommendations. This approach enables the system to provide personalized suggestions based on individual hand geometry.

In addition to the recommendation functionality, the system provides features that support service and business management. Mehndi artists can upload designs, manage bookings, and sell henna products, while customers can book artists, purchase products, and make secure online payments. The centralized database stores important information such as user details, design data, booking records, product information, and feedback, ensuring efficient data management.

The implementation results show that the system is capable of accurately detecting hand landmarks, extracting relevant features, and generating appropriate mehndi design recommendations. The integration of artificial intelligence with web-based services improves the efficiency of design selection and enhances user experience. Overall, the AI-Based Mehndi Design Recommendation System demonstrates the potential of combining computer vision and web technologies to modernize traditional art practices and provide a complete digital solution for users and artists.

9.2 Limitations

Although the AI-Based Mehndi Design Recommendation System performs effectively in automating mehndi design selection and providing integrated services such as booking and product sales, certain limitations still exist. These limitations arise mainly due to technical

constraints, dependency on image quality, and the simplicity of the recommendation approach used in the system.

One limitation of the system is its dependence on the dataset used for storing mehndi designs and their corresponding features. If the dataset does not contain sufficient variety in design styles or hand feature representations, the system may not provide accurate or diverse recommendations. A limited dataset can restrict the system's ability to match user hand features with the most suitable designs.

Another limitation is the dependence on image quality and input conditions. The system relies on uploaded hand images for feature extraction. Poor lighting, low-resolution images, incorrect hand positioning, or background noise may reduce the accuracy of hand landmark detection. This can lead to incorrect feature extraction and affect the quality of the recommended designs.

The system also uses a Euclidean distance-based algorithm, which is relatively simple and may not capture complex relationships between hand features and aesthetic design preferences. This may limit the accuracy of recommendations in cases where visual appeal or user preference plays a significant role beyond geometric measurements.

Additionally, the system requires adequate computational resources for image processing and feature extraction. While not as intensive as deep learning models, processing multiple images or handling large datasets may affect performance on low-end systems. Optimization may be required for large-scale deployment.

Since the platform operates as a web-based system, a stable internet connection is necessary for uploading images, generating recommendations, booking services, and completing online payments. Network issues or slow internet connectivity may affect system responsiveness and user experience.

These limitations highlight the need for continuous improvements such as expanding the dataset, integrating advanced machine learning models, improving image processing techniques, and enhancing system scalability to further improve the performance and reliability of the AI-Based Mehndi Design Recommendation System.

9.3 Future Scope

There are several opportunities to improve and extend the AI-Based Mehndi Design Recommendation System in the future. As artificial intelligence, computer vision, and web technologies continue to evolve, additional features can be integrated to enhance the functionality, accuracy, and user experience of the system.

One possible improvement is the integration of more advanced deep learning models for better recommendation accuracy. Future versions of the system could incorporate machine learning techniques such as neural networks or convolutional models to learn patterns from large datasets and provide more precise and personalized mehndi design suggestions.

Another enhancement could involve the implementation of a virtual try-on feature, where users can see how a selected mehndi design would appear on their hand in real time. This can be achieved using augmented reality (AR) technologies, allowing users to visualize designs before making a decision.

The system can also be expanded by incorporating user preference learning, where the platform tracks user choices, favorite designs, and booking history to provide more customized recommendations. This would improve personalization by considering both hand features and user interests.

Another future improvement could involve developing mobile applications so that users can access the platform through smartphones and tablets. A mobile app would make the system more accessible and convenient, allowing users to upload images, book artists, and make payments from anywhere.

In addition, the system could be integrated with cloud-based infrastructure to improve scalability and support a large number of users simultaneously. Cloud integration would also enhance data storage, processing speed, and system reliability.

The platform can also be extended by adding advanced analytics and business insights for mehndi artists. This would allow artists to track customer preferences, analyze sales trends, and improve their services based on data-driven insights.

Finally, future versions of the system could include multilingual support and cultural customization, allowing users from different regions to access the platform in their preferred language and receive design recommendations based on cultural styles and occasions.

Overall, the future scope of the AI-Based Mehndi Design Recommendation System highlights the potential for continuous improvement by integrating advanced technologies, enhancing personalization, and expanding platform accessibility to provide a more intelligent and user-friendly experience.

9.4 Applications

The **AI-Based Mehndi Design Recommendation System** has several practical applications in different domains related to beauty, art, and digital services. The system provides an intelligent platform for automating mehndi design selection, managing bookings, and supporting online product sales.

One important application is in **beauty salons and mehndi studios**, where the system can be used to recommend suitable mehndi designs to customers based on their hand features. This helps professionals provide better suggestions and improves customer satisfaction.

The system can also be used in **online mehndi service platforms**, where customers can browse designs, receive personalized recommendations, and book mehndi artists remotely. This makes the service more accessible and convenient, especially for users who prefer online booking.

Another application is in **freelance artist platforms**, where mehndi artists can showcase their work, manage bookings, and sell henna products. The system helps artists expand their reach and connect with a larger customer base through a digital platform.

The platform can also be useful for **event management services**, such as weddings, festivals, and cultural events, where mehndi services are commonly required. Event organizers can use the system to select designs and book artists efficiently.

Additionally, the system can be applied in **e-commerce platforms** for selling henna products and related accessories. Customers can browse products, compare prices, and make secure online payments through the system.

Finally, the system can be useful for **personal use**, where individuals can explore mehndi designs, receive recommendations, and experiment with different styles before making a final choice. The integration of artificial intelligence enhances the overall user experience by providing quick and personalized suggestions.

Overall, the AI-Based Mehndi Design Recommendation System has wide applications in the beauty and service industry, offering a modern and efficient solution for design recommendation, artist booking, and product sales.

CHAPTER 10

APPENDICES

```

import numpy as np
import random
from .models import Mehndi

def safe_float(value):
    """Convert None to 0"""
    if value is None:
        return 0.0
    return float(value)

def recommend_designs(features):
    # Convert user hand features to vector
    user_vector = np.array([
        safe_float(features.get("palm_width_ratio")),
        safe_float(features.get("palm_length_ratio")),
        safe_float(features.get("index_length")),
        safe_float(features.get("middle_length")),
        safe_float(features.get("ring_length")),
        safe_float(features.get("pinky_length")),
        safe_float(features.get("thumb_length")),
    ])

    designs = Mehndi.objects.all()

    scored = []

    for design in designs:
        # Convert database features to vector
        db_vector = np.array([
            safe_float(design.palm_width_ratio),
            safe_float(design.palm_length_ratio),
            safe_float(design.index_length),
            safe_float(design.middle_length),
            safe_float(design.ring_length),
            safe_float(design.pinky_length),
            safe_float(design.thumb_length),
        ])

        # Euclidean Distance
        distance = np.linalg.norm(user_vector - db_vector)

        scored.append((distance, design))

    # Sort by closest match
    scored.sort(key=lambda x: x[0])

    # If distances are almost same, shuffle results
    top = scored[:10]
    random.shuffle(top)

    # Return top 5
    return [x[1] for x in top[:5]]

```

```

import cv2
import mediapipe as mp
import math

mp_hands = mp.solutions.hands

def distance(p1, p2):
    return math.sqrt((p1.x - p2.x) ** 2 + (p1.y - p2.y) ** 2)

def extract_hand_features(image_path):
    image = cv2.imread(image_path)
    image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

    with mp_hands.Hands(static_image_mode=True) as hands:
        results = hands.process(image_rgb)

        if not results.multi_hand_landmarks:
            return None

        landmarks = results.multi_hand_landmarks[0].landmark

        palm_width = distance(landmarks[5], landmarks[17])
        palm_length = distance(landmarks[0], landmarks[9])

        index_length = distance(landmarks[5], landmarks[8])
        middle_length = distance(landmarks[9], landmarks[12])
        ring_length = distance(landmarks[13], landmarks[16])
        pinky_length = distance(landmarks[17], landmarks[20])

        thumb_length = distance(landmarks[2], landmarks[4])

from django.contrib import admin
from django.urls import path
from django.conf.urls.static import static
from myapp import views

from mehandi_project import settings

urlpatterns = [

    path('index/', views.index),
    path('forgot_password_get/', views.forgot_password_get),
    path('forgot_password/', views.forgot_password),
    path('login_get/', views.login_get),
    path('login_post/', views.login_post),
    path('admin_home/', views.admin_home),
    path('admin_view_design/<id>', views.admin_view_design),
    path('admin_view_user/', views.admin_view_user),
    path('admin_reject_artist/<id>', views.admin_reject_artist),
    path('admin_accept_artist/<id>', views.admin_accept_artist),
    path('admin_view_artist/', views.admin_view_artist),
    path('admin_view_complaints/', views.admin_view_complaints),
    path('admin_send_reply/', views.admin_send_reply),
    path('admin_view_feedback/', views.admin_view_feedback),
    path('admin_accept_product/<id>', views.admin_accept_product),
    path('admin_reject_product/<id>', views.admin_reject_product),
    path('admin_view_product/<id>', views.admin_view_product),
    path('admin_add_recommend_get/', views.admin_add_recommend_get),
    path('admin_add_recommend_post/', views.admin_add_recommend_post),
    path('admin_edit_design_get/<id>', views.admin_edit_design_get),
    path('admin_edit_recommend_post/', views.admin_edit_design_post),
    path('admin_delete_design/<id>', views.admin_delete_design),

```

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>MehndiVerse · edit recommendation</title>
  <!-- Font Awesome 6 -->
  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/css/all.min.css">
  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
    }

    body {
      background: #f8efe6;
      font-family: 'Inter', 'Segoe UI', system-ui, sans-serif;
      color: #2e1c12;
      display: flex;
      flex-direction: column;
      min-height: 100vh;
    }

    /* main content */
    .edit-container {
      flex: 1;
      max-width: 900px;
      width: 100%;
      margin: 30px auto 40px;
      padding: 0 25px;
    }
  
```

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>MehndiVerse · artist login</title>
  <!-- Font Awesome 6 -->
  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/css/all.min.css">
  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
    }

    body {
      font-family: 'Outfit', 'Segoe UI', system-ui, sans-serif;
      background: #ffff4;
      color: #2e1c12;
      min-height: 100vh;
      display: flex;
      align-items: center;
      justify-content: center;
      padding: 20px;
      /* USING YOUR SPECIFIED BACKGROUND IMAGE */
      background: linear-gradient(145deg, rgba(250, 240, 230, 0.8), rgba(230, 200, 175, 0.75)),
        url('https://mehndidesignlist.com/wp-content/uploads/2025/06/royal-dulhan-mehndi-design.jpg');
      background-size: cover;
      background-position: center;
      background-attachment: fixed;
    }
  
```

CHAPTER 11

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